

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

NET206

Amazon Route 53: What's in a domain name? A lot!

Kimberley Clements

Sr. Solutions Architect
AWS

Steve Seymour

Sr. Principal Solutions Architect
AWS



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

A few ground rules ...

1. It's always DNS
2. 200 Level – overview, basic functions, use cases
3. Domain transfers, registration, DNSSEC, permissions
4. The word Route



The end



Agenda



What is a
domain name?



What is
Route 53?



Routing
policies



DNS
security



Putting it all
together

Agenda



What is a
domain name?



What is
Route 53?



Routing
policies



DNS
security



Putting it all
together

What is a domain name?



What is DNS?



web1.unicornpacket.com



192.0.2.11

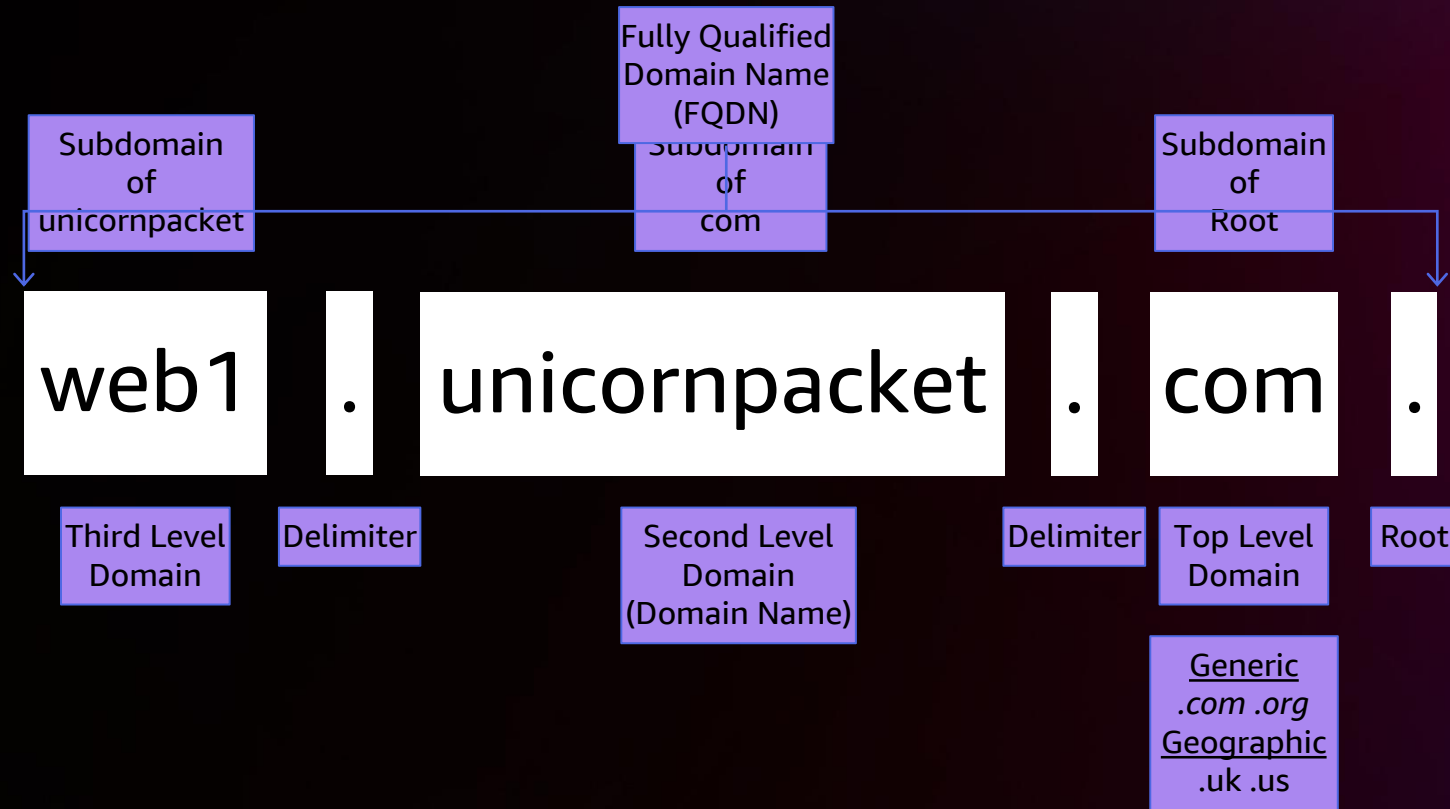
OR

2001:db8:ec2:1::1

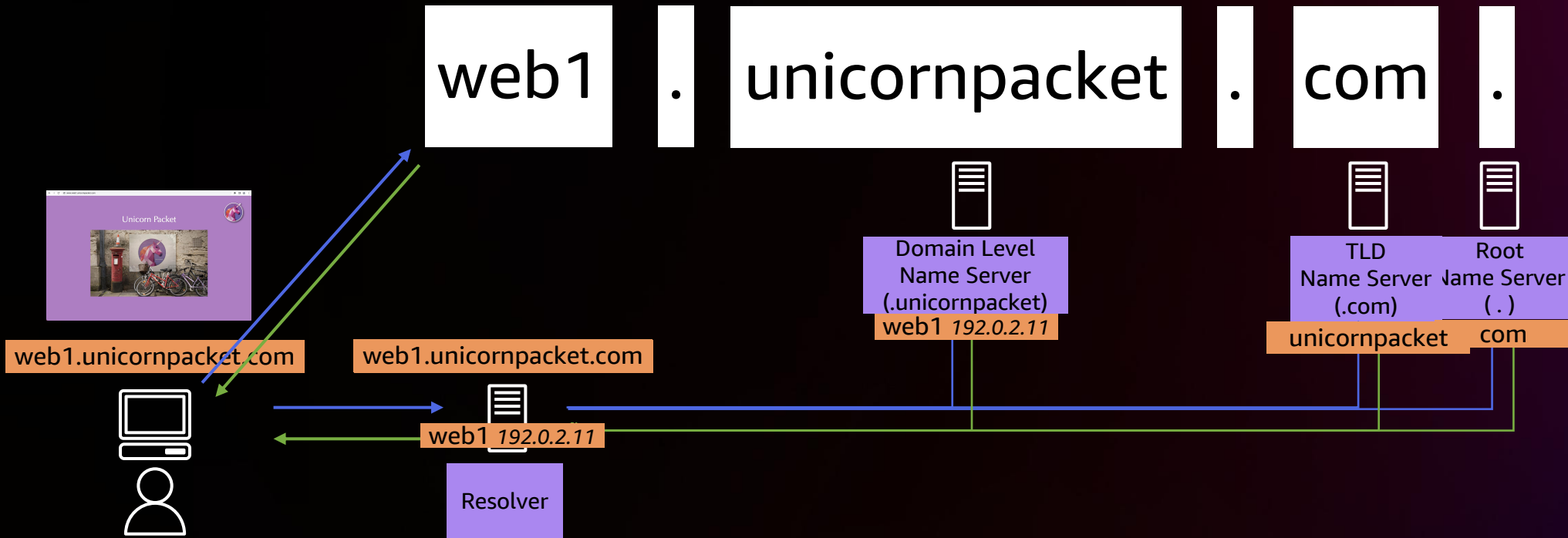
What is a domain name?

web1 . unicornpacket . com .

What is a domain name?



How to resolve that domain name?



Common record types

Record	Description
A	Routes traffic to an IPv4 address
AAAA	Routes traffic to an IPv6 address
CNAME	Routes traffic to another domain name
MX	Specifies mail servers
TXT	Used to verify email senders and application specific values
PTR	Maps an IP address to a domain name
SRV	Application-specific values that identify servers

Alias

A type of record that you can create with [Amazon Route 53](#) to route traffic to AWS resources such as Amazon CloudFront distributions and Amazon S3 buckets

 Alias

Route traffic to [Info](#)

Choose endpoint

Alias to another record in this hosted zone

Alias to API Gateway API

Alias to AppRunner service

Alias to AppSync Domain Name

Alias to CloudFront distribution

Alias to Elastic Beanstalk environment

Alias to Application and Classic Load Balancer

Alias to Network Load Balancer

Alias to Global Accelerator

Alias to S3 website endpoint

Alias to VPC endpoint

Agenda



What is a
domain name?



What is
Route 53?



Routing
policies



DNS
security



Putting it all
together

Agenda



What is a
domain name?



What is
Route 53?



Routing
policies



DNS
security



Putting it all
together

What is Route 53



Route 53

- Domain name registration services
- Public authoritative DNS
- Private hosted zones
- Route 53 Resolver
- Integration with on-premises DNS environments
- DNS Firewall



Route 53

- Domain name registration services
- Public authoritative DNS



Route 53

- Domain name registration services
- Public Authoritative DNS



Domain name registration services



1. Registration of ownership information - "whois"
 2. Addition of a delegation record into the parent zone (gTLD or ccTLD)
- Supported top-level domain names
 - 272 generic top-level domains (gTLD) – .com .net .org etc.
 - 49 top-level country domains (ccTLD) – .uk .au .ca .de etc.

Therefore

"unicornpacket.com" is owned by Unicorn Co. (privacy enabled)

"unicornpacket" has been entered into the **".com"** gTLD zone file

- ```
Domain Name: unicornpacket.com
Registry Domain ID: 2729736169_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.registrar.amazon.com
Registrar URL: https://registrar.amazon.com
Updated Date: 2022-10-04T15:19:03Z
Creation Date: 2022-10-04T15:08:14Z
Registrar Registration Expiration Date: 2023-10-04T15:08:14Z
Registrar: Amazon Registrar, Inc.
Registrar IANA ID: 468
Registrar Abuse Contact Email: abuse@amazonaws.com
Registrar Abuse Contact Phone: +1.2067406200
Domain Status: ok https://icann.org/epp#ok
Registry Registrant ID: Not Available From Registry
Registrant Name: On behalf of unicornpacket.com owner
```

- |                           |        |    |       |                                                                                                                                                                                                                                                                                |
|---------------------------|--------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unicornpackersmovers.com. | 172800 | IN | NS    | ns1.dns-parking.com.                                                                                                                                                                                                                                                           |
| unicornpackersmovers.com. | 172800 | IN | NS    | ns2.dns-parking.com.                                                                                                                                                                                                                                                           |
| unicornpacket.com.        | 172800 | IN | NS    | ns-1500.awsdns-59.org.                                                                                                                                                                                                                                                         |
| unicornpacket.com.        | 172800 | IN | NS    | ns-1762.awsdns-28.co.uk.                                                                                                                                                                                                                                                       |
| unicornpacket.com.        | 172800 | IN | NS    | ns-254.awsdns-31.com.                                                                                                                                                                                                                                                          |
| unicornpacket.com.        | 172800 | IN | NS    | ns-563.awsdns-06.net.                                                                                                                                                                                                                                                          |
| unicornpacket.com.        | 86400  | IN | DS    | 34962 13 2 7EE5C38BFCEFD98BB71227F2BCFAD11D5ADB8BBCA4EFADA2D34CA6FCDBA2FA                                                                                                                                                                                                      |
| unicornpacket.com.        | 86400  | IN | RRSIG | DS 8 2 86400 20221018212241 20221011201241 32298 com. pW0UW74puUgwrCaTxHficW0x92uQV0QjjSVj0Ev4FhYkjQ9NGFzHQ832rNcqEIuLK2ESik5n0y3VKvVwj2m5Hjwds5k65IFhNpUfS/z4Xg6wTD7SkZouQ/jhrNI4maF/6GtKjGDGA5tSgh7reAN6vqUMiprCgaf0zQeIWmTai28gjWrMN2MLcSw3C1z7rS0LFnIVy0msx3zGTbcpN4SE3w== |
| unicornpacks.com.         | 172800 | IN | NS    | nsg1.namebrightdns.com.                                                                                                                                                                                                                                                        |
| unicornpacks.com.         | 172800 | IN | NS    | nsg2.namebrightdns.com.                                                                                                                                                                                                                                                        |
| unicornpacnmov.com.       | 172800 | IN | NS    | ns.uttlsolutions.net.                                                                                                                                                                                                                                                          |
| unicornpacnmov.com.       | 172800 | IN | NS    | ns1.uttlsolutions.com.                                                                                                                                                                                                                                                         |

# Domain names in Route 53



- Register a new domain name

**1: Domain Search**  
  
**2: Contact Details**  
  
**3: Verify & Purchase**

## Choose a domain name

### Availability for 'unicornpackets.com'

| Domain Name        | Status                | Price /1 Year | Action                                     |
|--------------------|-----------------------|---------------|--------------------------------------------|
| unicornpackets.com | ✓ Available - In Cart | \$12.00       | <input type="button" value="Add to cart"/> |

**1: Domain Search**  
  
**2: Contact Details**  
  
**3: Verify & Purchase**

## Contact Details for Your 1 Domain

Enter the details for your Registrant, Administrative and Technical contacts below. All fields are required unless specified otherwise. [Learn more.](#)

My Registrant, Administrative and Technical Contacts are all the same: ☒ Yes ☐ No

### Registrant Contact

Contact Type

Company

First Name

Domain

Last Name

Registrar

Organization

Unicorn Packets

Email

reinvent-net206@

Phone

+44 - 795

Address 1

60 Holborn Viaduct

Address 2

London

Country

United Kingdom

State

State not required

City

London

Postal/Zip Code

EC1A 2FD

Privacy Protection

When the contact type is Company:

- Privacy protection hides some contact details for .com domains.

☒ Enable ☐ Disable

**1: Domain Search**  
  
**2: Contact Details**  
  
**3: Verify & Purchase**

## Check your contact details

Confirm that the following contact information is correct. When you complete your purchase, we'll use this information for all of the domains in your shopping cart.

**Registrant Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

**Administrative Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

**Technical Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

### Managing DNS for Your New Domain

To make it easier for you to use Route 53 as the DNS service for your new domain, we'll automatically create a hosted zone. That's where you store information about how to route traffic for your domain, for example, to an Amazon EC2 instance. If you won't use your domain right now, you can delete the hosted zone. If you will use your domain, Route 53 charges for the hosted zone and for the DNS queries that we receive for your domain. For more information, see [Amazon Route 53 Pricing](#).

### Do you want to automatically renew your domain?

When you register a domain name, you own it for a year. If you don't renew your domain name registration, it expires and someone else can register the domain name. To ensure that you can keep your domain name, you can choose to renew it automatically every year. The cost of renewing your domain name is billed to your AWS account. You can enable or disable automatic renewal at any time using the Route 53 console. For more information, see [Renewing Registration for a Domain](#).

☒ Enable ☐ Disable

### Terms and Conditions

Amazon Route 53 enables you to register and transfer domain names using your AWS account. However, AWS is not a domain name registrar, so we use registrar associates to perform registration and transfer services. When you purchase domain names through AWS, you are registering your domain with one of our registrar associates. The registrar for your domain will periodically contact the registrant contact that you specified to verify the contact details and renew registration.

☒ I have read and agree to the [AWS Domain Name Registration Agreement](#)

### Verify the Email Address for the Registrant Contact

Status: ✓ reinvent-net206@

is verified

© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

# Domain names in Route 53



- Register a new domain name

**1: Domain Search**

2: Contact Details

3: Verify & Purchase

### Choose a domain name

.com - \$12.00 Check

**Availability for 'unicornpackets.com'**

| Domain Name        | Status                | Price /1 Year | Action                       |
|--------------------|-----------------------|---------------|------------------------------|
| unicornpackets.com | ✓ Available - In Cart | \$12.00       | <button>Add to cart</button> |

1: Domain Search

2: Contact Details

3: Verify & Purchase

### Contact Details for Your 1 Domain

Enter the details for your Registrant, Administrative and Technical contacts below. All fields are required unless specified otherwise. [Learn more.](#)

My Registrant, Administrative and Technical Contacts are all the same: ☒ Yes ☐ No

**Registrant Contact**

**Contact Type**

Company

**First Name**

Domain

**Last Name**

Registrar

**Organization**

Unicorn Packets

**Email**

reinvent-net206@

**Phone**

+44 - 795

**Address 1**

60 Holborn Viaduct

**Address 2**

London

**Country**

United Kingdom

**State**

State not required

**City**

London

**Postal/Zip Code**

EC1A 2FD

**Privacy Protection**

When the contact type is Company:

- Privacy protection hides some contact details for .com domains.

☒ Enable ☐ Disable

1: Domain Search

2: Contact Details

3: Verify & Purchase

### Check your contact details

Confirm that the following contact information is correct. When you complete your purchase, we'll use this information for all of the domains in your shopping cart.

**Registrant Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

**Administrative Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

**Technical Contact**

Domain Registrar  
Unicorn Packets  
reinvent-net206@

60 Holborn Viaduct London  
EC1A 2FD  
GB  
Privacy protected

### Managing DNS for Your New Domain

To make it easier for you to use Route 53 as the DNS service for your new domain, we'll automatically create a hosted zone. That's where you store information about how to route traffic for your domain, for example, to an Amazon EC2 instance. If you won't use your domain right now, you can delete the hosted zone. If you will use your domain, Route 53 charges for the hosted zone and for the DNS queries that we receive for your domain. For more information, see [Amazon Route 53 Pricing](#).

### Do you want to automatically renew your domain?

When you register a domain name, you own it for a year. If you don't renew your domain name registration, it expires and someone else can register the domain name. To ensure that you can keep your domain name, you can choose to renew it automatically every year. The cost of renewing your domain name is billed to your AWS account. You can enable or disable automatic renewal at any time using the Route 53 console. For more information, see [Renewing Registration for a Domain](#).

☒ Enable ☐ Disable

### Terms and Conditions

Amazon Route 53 enables you to register and transfer domain names using your AWS account. However, AWS is not a domain name registrar, so we use registrar associates to perform registration and transfer services. When you purchase domain names through AWS, you are registering your domain with one of our registrar associates. The registrar for your domain will periodically contact the registrant contact that you specified to verify the contact details and renew registration.

☒ I have read and agree to the [AWS Domain Name Registration Agreement](#)

### Verify the Email Address for the Registrant Contact

Status: ✓ reinvent-net206@emealab.net is verified

Cancel Back Complete Order

© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

# Domain names in Route 53



- Register a new domain name

Registered domains > unicornpackets.com

[Edit contacts](#) [Manage DNS](#) [Delete domain](#)

|                            |                                       |                                  |                                                                        |                        |                                                                                                                                              |
|----------------------------|---------------------------------------|----------------------------------|------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Domain</b>              | unicornpackets.com                    | <b>Transfer lock</b> ⓘ           | Disabled ( <a href="#">enable</a> )                                    | <b>Name servers</b> ⓘ  | ns-526.awsdns-01.net<br>ns-439.awsdns-54.com<br>ns-1152.awsdns-16.org<br>ns-1562.awsdns-03.co.uk<br><a href="#">Add or edit name servers</a> |
| <b>Registration date</b> ⓘ | 2022-10-14                            | <b>Authorization code</b> ⓘ      | <a href="#">Get code</a>                                               |                        |                                                                                                                                              |
| <b>Expiration date</b> ⓘ   | 2023-10-14 ( <a href="#">extend</a> ) | <b>Domain name status code</b> ⓘ | addPeriod<br>ok                                                        | <b>DNSSEC status</b> ⓘ | Disabled<br><a href="#">Manage keys</a>                                                                                                      |
| <b>Auto renew</b> ⓘ        | Enabled ( <a href="#">disable</a> )   | <b>Tag</b> ⓘ                     | View and manage tags for your domains using <a href="#">Tag editor</a> |                        |                                                                                                                                              |

|                                                     |                                                     |                                                     |
|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| <b>Registrant contact</b> <span>Verified</span>     | <b>Administrative contact</b>                       | <b>Technical contact</b>                            |
| Domain Registrar<br>reinvent-net206@                | Domain Registrar<br>reinvent-net206@                | Domain Registrar<br>reinvent-net206@                |
| 60 Holborn Viaduct, London<br>London EC1A 2FD<br>GB | 60 Holborn Viaduct, London<br>London EC1A 2FD<br>GB | 60 Holborn Viaduct, London<br>London EC1A 2FD<br>GB |

"To make it easier for you to use Route 53 as the DNS service for your new domain, we'll automatically create a hosted zone"

# Domain names in Route 53



- Register a new Domain name
- Transfer an existing supported domain name

1: Select Domain

2: Domain Options

3: Contact Details

4: Review & Purchase

## Transfer Domain to Route 53

You can transfer registration for one or more domains from another registrar to Route 53. Before you continue, do the following:

- Confirm that the domain is transferable. See [Transfer requirements for top-level domains](#).
- For each domain that you want to transfer, perform the first four steps of [Transferring registration for a domain to Route 53](#).

To transfer up to five domains, you can enter each domain name below.  
To transfer more than five domains, you can use the [Transfer multiple domains to Route 53 page](#).

Check

unicornpacket.com can be transferred to Route 53 (\$12.00)

To transfer the domain to Route 53 please ensure the following:

- Disable domain privacy for the domain so we can get your contact information.
- Ensure the domain's registrant email at WHOIS is valid and reachable.
- Get an authorization code for the domain (*Not required for certain TLDs*).

Add to Cart

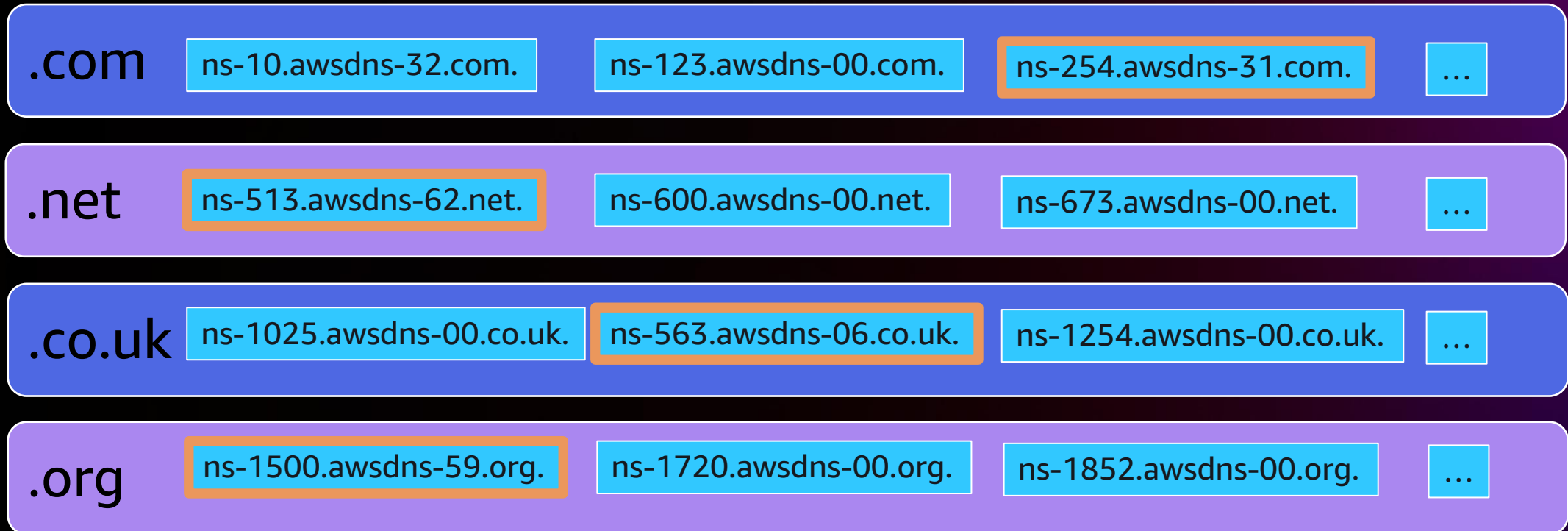
# Route 53

- Domain Name Registration Services
- Public authoritative DNS



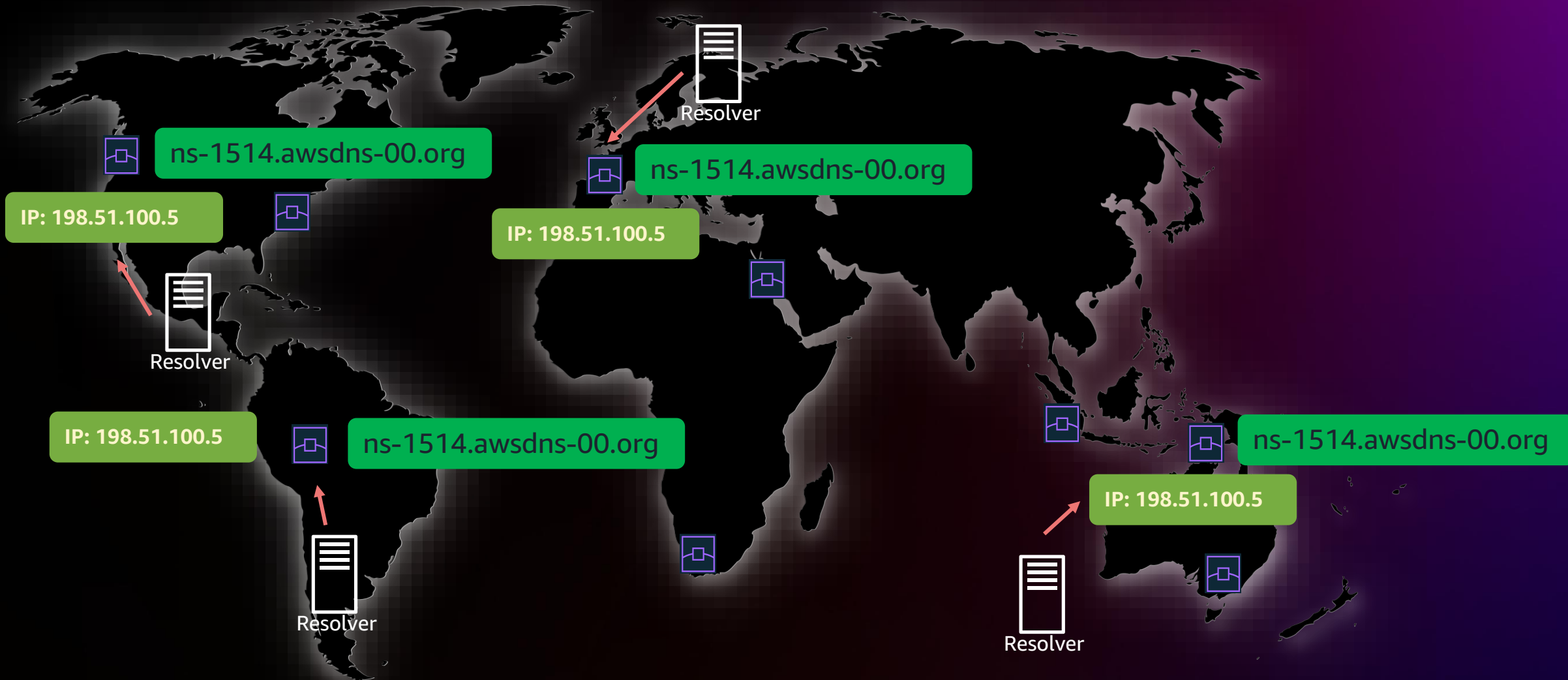


# Route 53 architecture – name server stripes



unicornpacket.com

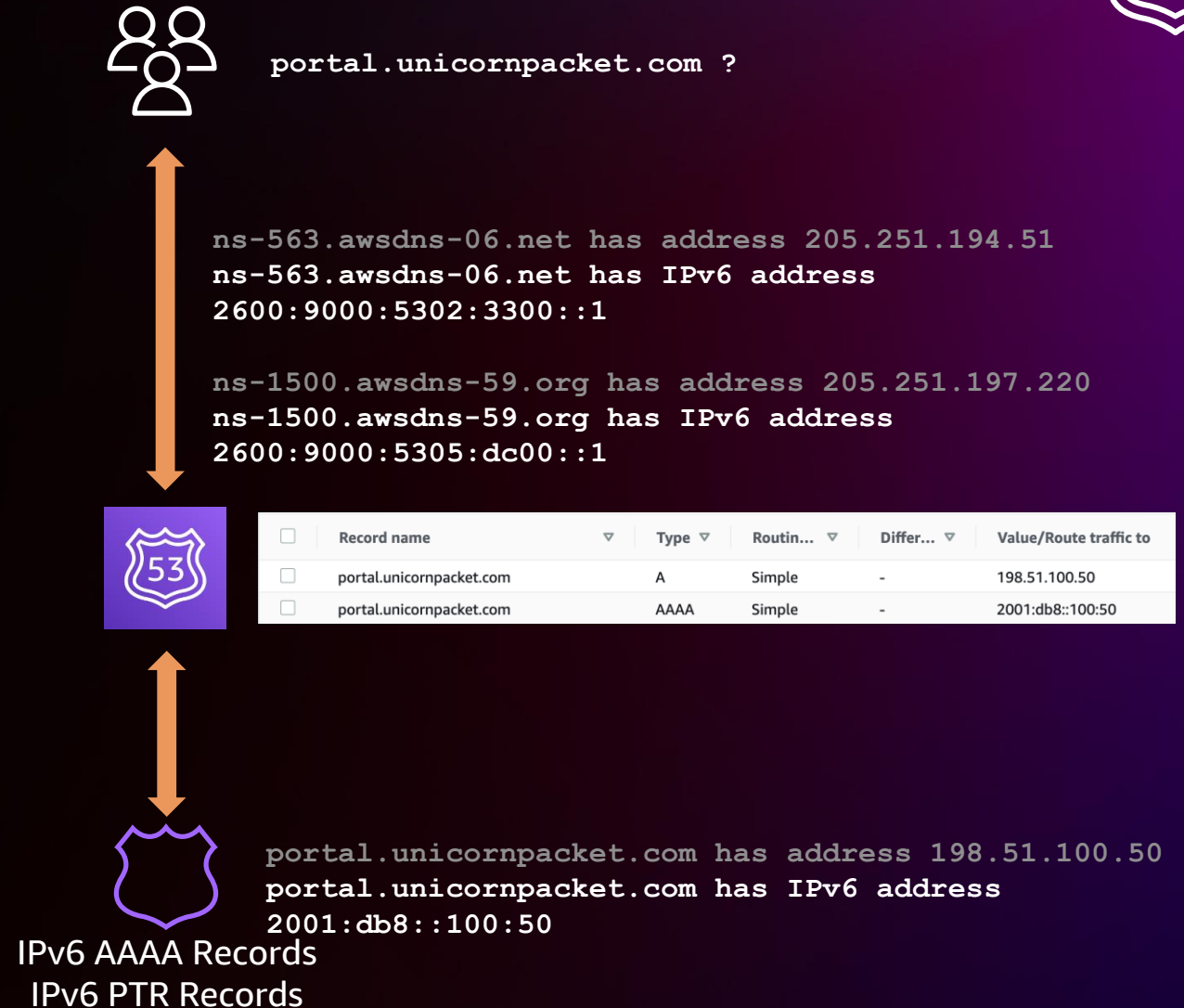
# Route 53 architecture – anycast



# Route 53 IPv6 support



- End-to-end DNS resolution over IPv6
- Support for IPv6 Forward (AAAA) and reverse (PTR) DNS records
- Route 53 health checks support monitoring IPv6 endpoints

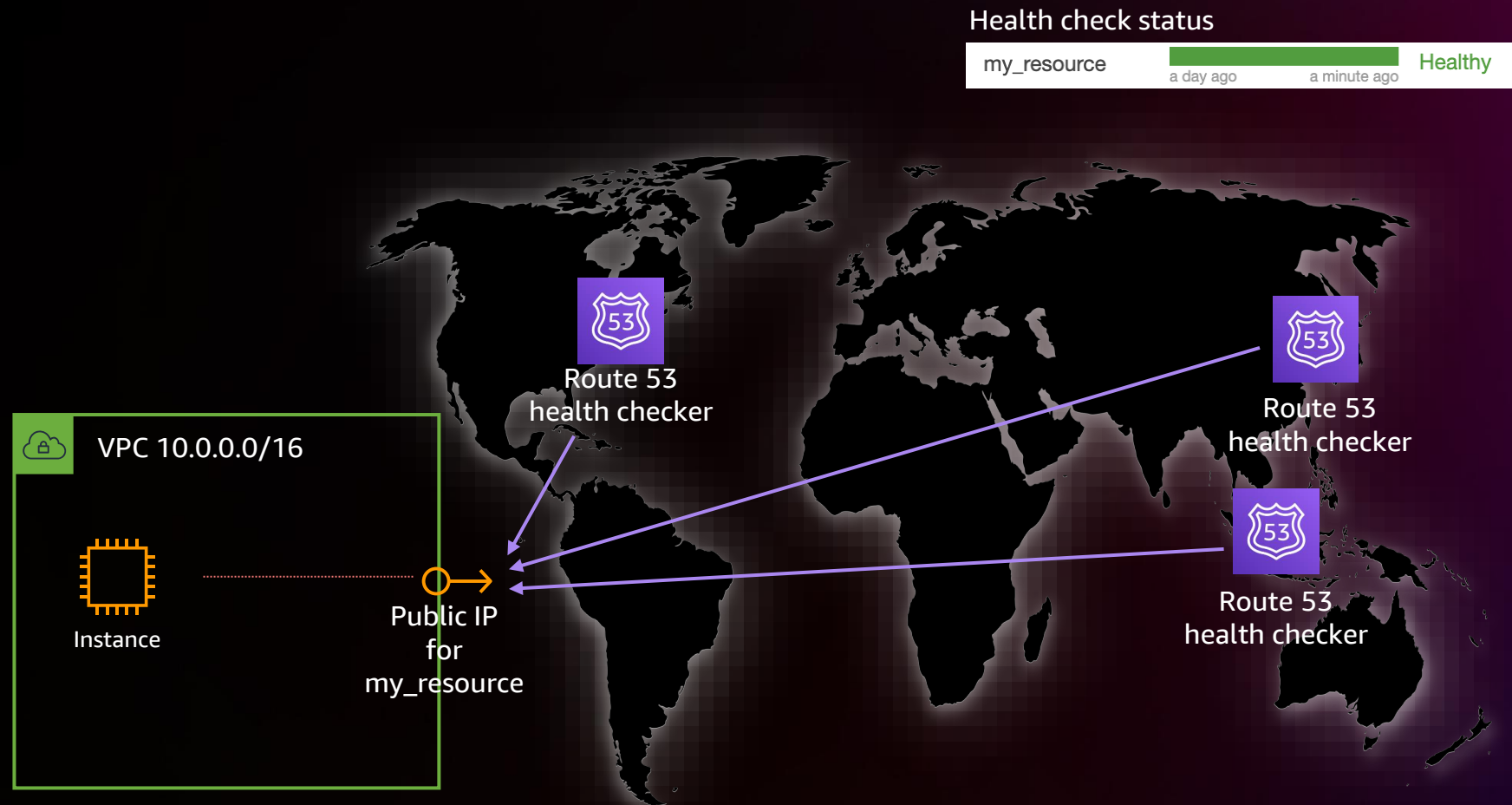


# Health checks

- The health of a specified resource
- The status of other health checks
- The status of an Amazon CloudWatch alarm



# Health checks – endpoint



# Health checks – status of other checks

my\_resource 1  Healthy  
a day ago a minute ago

my\_resource 2  Healthy  
a day ago a minute ago

my\_resource 3  Unhealthy  
a day ago now

Calculated health check status



calculated health check  Healthy  
19 minutes ago 4 minutes ago

## Monitor other health checks (calculated health check)

The health of this health check depends on the status of the following health checks:

Health checks to monitor

my\_resource\_1

×

ⓘ

my\_resource\_2

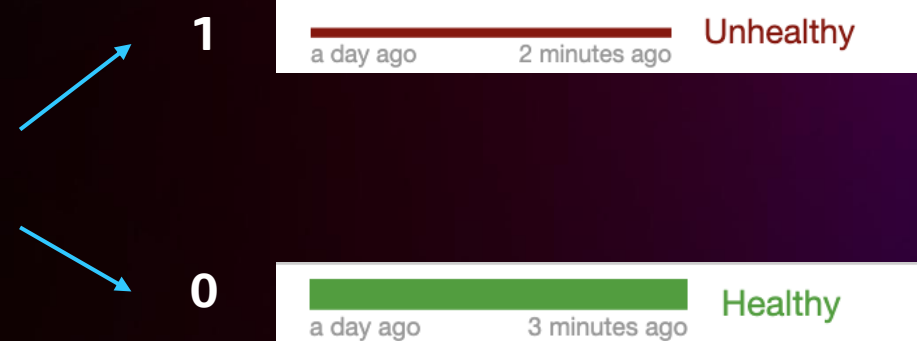
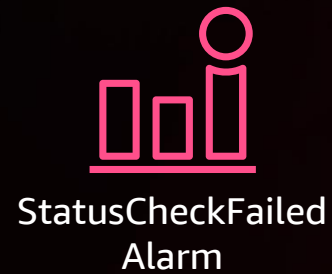
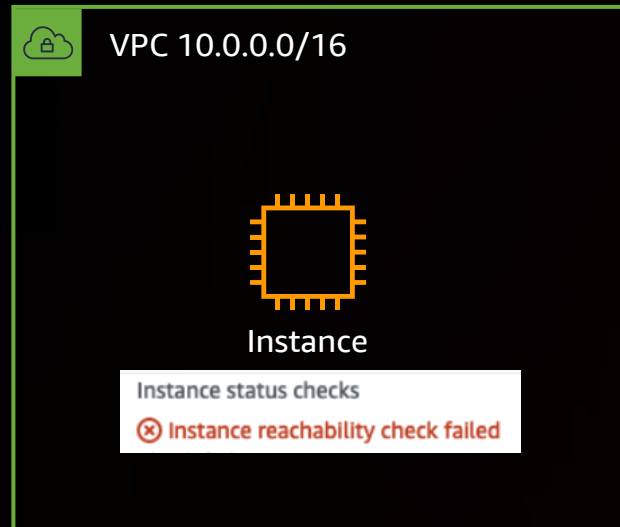
×

my\_resource\_3

×

- Report healthy when
- ☒ at least  of 3 selected health checks are healthy ⓘ
  - ☐ all health checks are healthy (AND)
  - ☐ one or more health checks are healthy (OR)

# Health checks – CloudWatch alarm



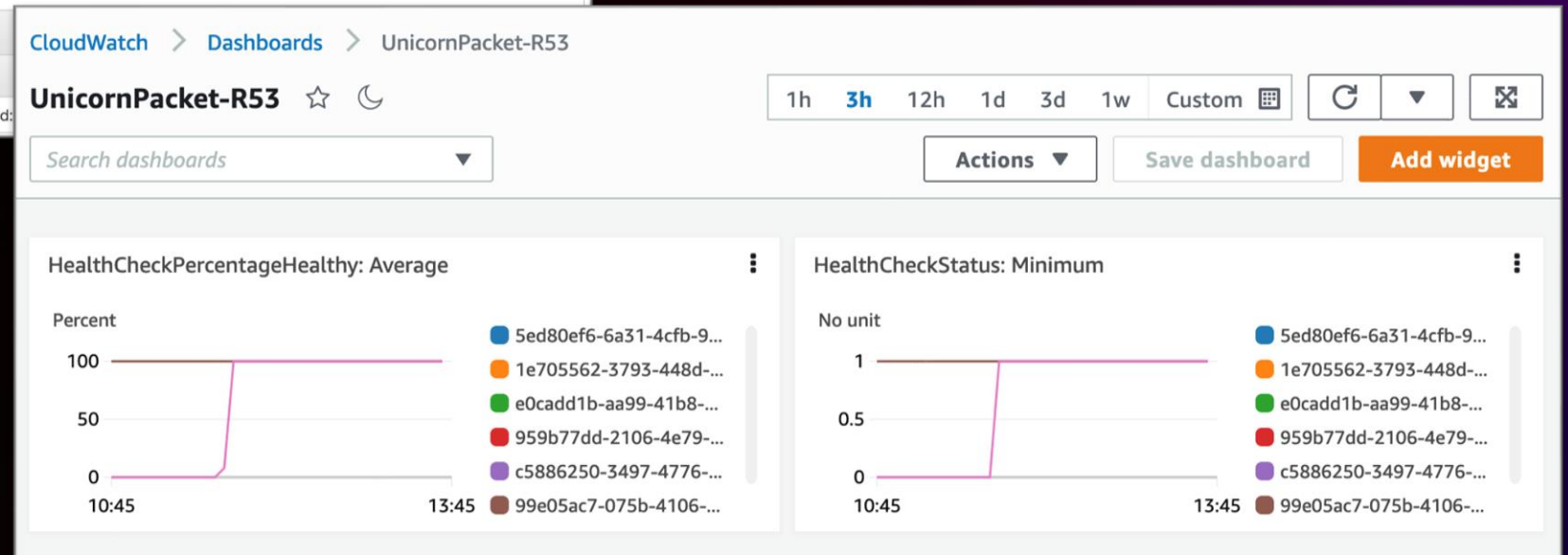
# CloudWatch metrics and dashboards



## DNS queries



## Health check status





# Query logging



Route 53 > Hosted zones > unicornpacket.com > Configure query logging

## Configure query logging [Info](#)

You can configure Amazon Route 53 to log information about the queries that Route 53 receives, such as the domain or subdomain that was requested, the date and time of the query, and the DNS record type (such as A or AAAA).

### Log group [Info](#)

Specify the CloudWatch Logs log group where you want Route 53 to save DNS queries for records in this hosted zone.

**Log group**  
You can choose the name of an existing log group or choose to create a new log group.

Create log group ▼ 

New log group name

unicornpacket-query-logs

The log group can have up to 512 characters. Valid characters: a-z, A-Z, 0-9, and . \_ / # - (hyphen)

**Missing permission**

Route 53 needs permission from a resource policy to publish logs to a CloudWatch Logs log group. No existing resource policies grant the required permissions.

► Details

**Grant permissions**

► Permissions - optional

Cancel

**Create**

# Query logging



CloudWatch > Log groups > unicornpacket-query-logs > Z08033393PJJ58GDGJSFY/NRT12-UU1

### Log events

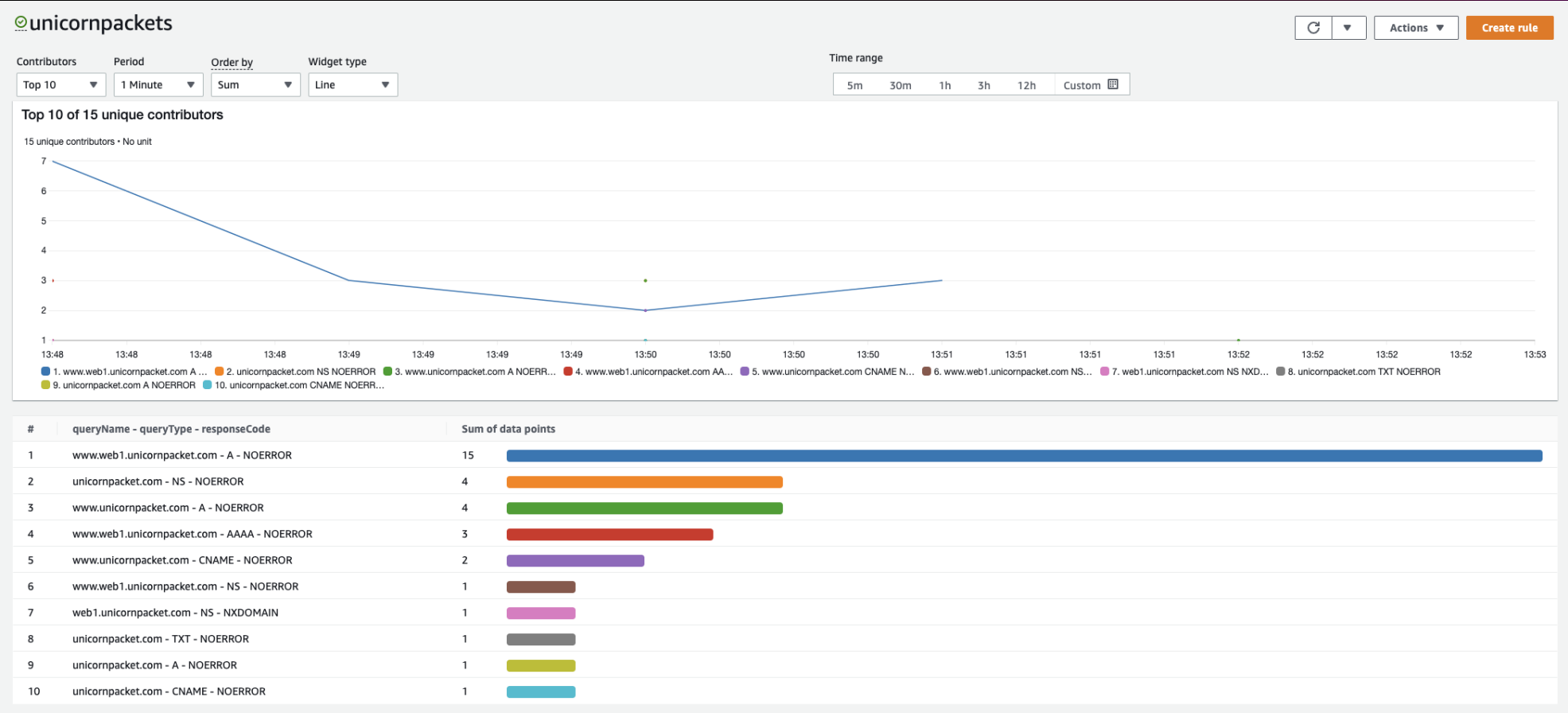
You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

☐ View as text **Actions**

1m 30m 1h 12h Custom

| ▶                                                                         | Timestamp                     | Message                                                                                                                                                                                                                    |                                     |
|---------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| No older events at this moment. <a href="#">Retry</a>                     |                               |                                                                                                                                                                                                                            |                                     |
| ▼                                                                         | 2022-10-12T10:24:40.000+01:00 | 1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com DNSKEY NOERROR UDP NRT12-UU1 35.72.37.67 -<br>1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com DNSKEY NOERROR UDP NRT12-UU1 35.72.37.67   | <input type="button" value="Copy"/> |
| ▼                                                                         | 2022-10-12T10:24:40.000+01:00 | 1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com A NOERROR UDP NRT12-UU1 35.72.37.74 -<br>1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com A NOERROR UDP NRT12-UU1 35.72.37.74 -           | <input type="button" value="Copy"/> |
| ▼                                                                         | 2022-10-12T10:24:40.000+01:00 | 1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com DNSKEY NOERROR UDP NRT12-UU1 35.72.37.74 -<br>1.0 2022-10-12T09:24:40Z Z08033393PJJ58GDGJSFY unicornpacket.com DNSKEY NOERROR UDP NRT12-UU1 35.72.37.74 - | <input type="button" value="Copy"/> |
| No newer events at this moment. Auto retry paused. <a href="#">Resume</a> |                               |                                                                                                                                                                                                                            |                                     |

# Query logging

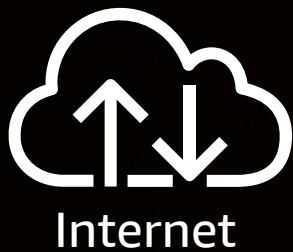


# Hosted zones: Public and private

# What is the difference?

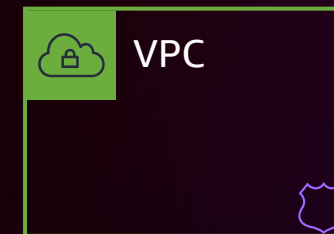
## Public hosted zones

- Route to Internet facing resources
- Resolve from the Internet
- Provides Internet reachable nameservers
- Can have be delegated to from parent zones
- Supports delegation of sub-domains
- Global routing policies



## Private hosted zones

- Route to VPC resources
- Resolve from inside the VPC (or private on-premises network)
- Reachable using forwarding rules and endpoints
- Does not support delegation
- Integrated with Route 53 Resolver (VPC+2)



# Public hosted zone

- Name servers/delegation sets
- Resource record sets
- TTL & caching
- DNSSEC configuration
- Importing zone file contents
- Metrics & query logging
- AWS specific record types
  - Alias
  - Routing policies

Route 53 > Hosted zones

### Hosted zones (3)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

 [View details](#) [Edit](#) [Delete](#) [Create hosted zone](#)



|                       | Domain name ▾                      | Type ▾ | Created by ▾ | Record count ▾ |
|-----------------------|------------------------------------|--------|--------------|----------------|
| <input type="radio"/> | <a href="#">unicornpacket.com</a>  | Public | Route 53     | 13             |
| <input type="radio"/> | <a href="#">unicornpackets.com</a> | Public | Route 53     | 2              |
| <input type="radio"/> | <a href="#">unicorndns.net</a>     | Public | Route 53     | 2              |

[Public](#) [unicornpacket.com](#) [Info](#) [Delete zone](#) [Test record](#) [View query logging configuration](#)

### Hosted zone details

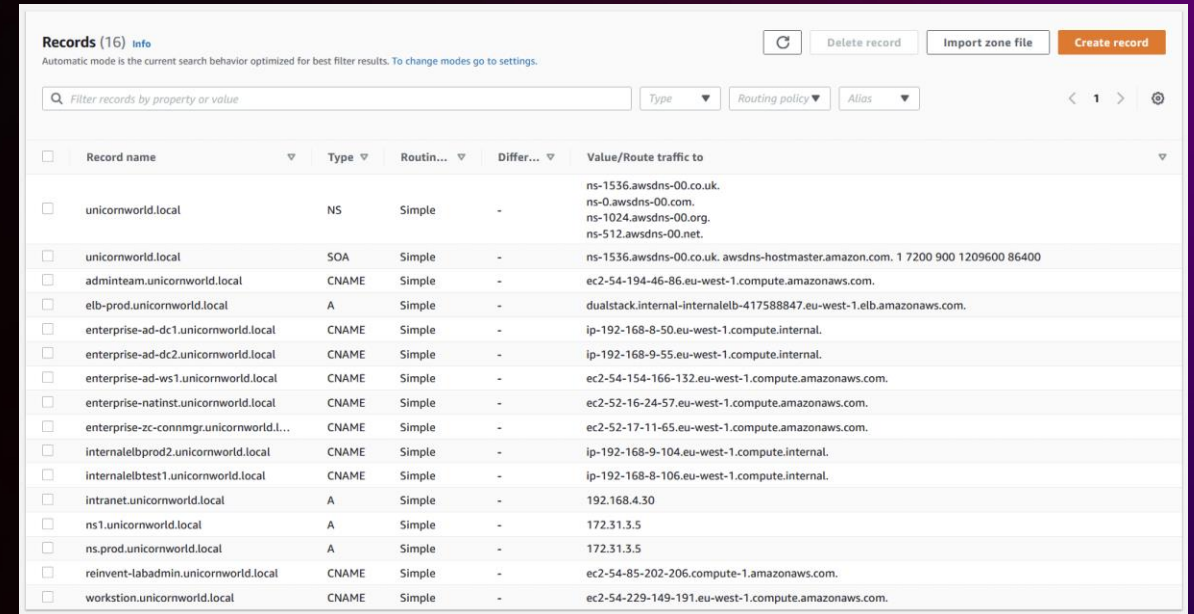
[Edit hosted zone](#)

|                                                     |                                         |                                                                                                                  |
|-----------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Hosted zone name<br>unicornpacket.com               | Query log<br>unicornpacket-query-logs:* | Name servers<br>ns-254.awsdns-31.com<br>ns-1762.awsdns-28.co.uk<br>ns-1500.awsdns-59.org<br>ns-563.awsdns-06.net |
| Hosted zone ID<br>Z08033393PJS8GDGJSFY              | Type<br>Public hosted zone              |                                                                                                                  |
| Description<br>Production Domain for Unicorn Packet | Record count<br>13                      |                                                                                                                  |



# Private hosted zones

- Takes priority in a VPC
- Can be part of a split view setup
- Shared cross-account
- Shared between VPCs

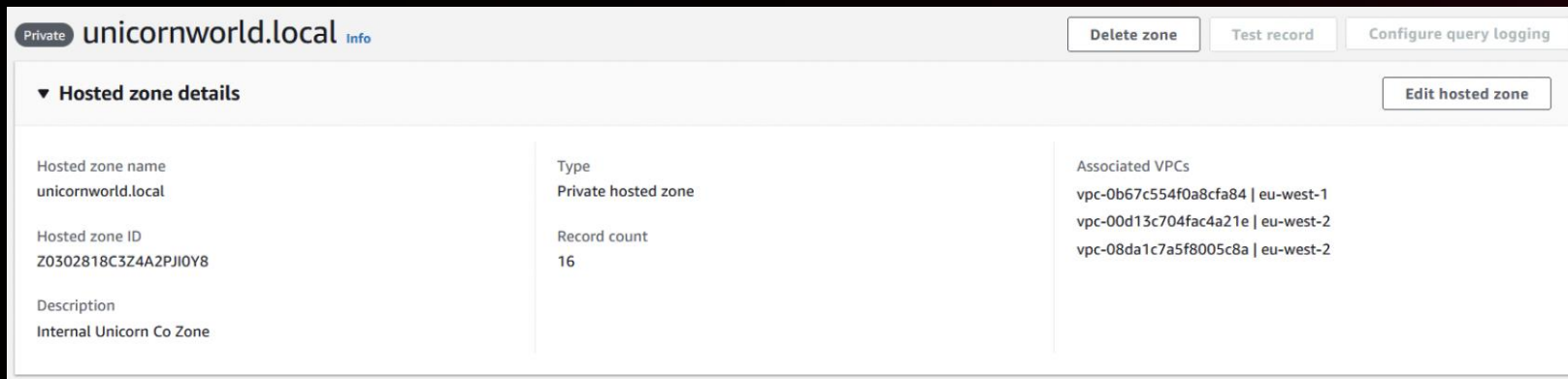


Records (16) info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

| Record name                              | Type  | Routin... | Differ... | Value/Route traffic to                                                                             |
|------------------------------------------|-------|-----------|-----------|----------------------------------------------------------------------------------------------------|
| unicornworld.local                       | NS    | Simple    | -         | ns-1536.awsdns-00.co.uk.<br>ns-0.awsdns-00.com.<br>ns-1024.awsdns-00.org.<br>ns-512.awsdns-00.net. |
| unicornworld.local                       | SOA   | Simple    | -         | ns-1536.awsdns-00.co.uk. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400                    |
| adminteam.unicornworld.local             | CNAME | Simple    | -         | ec2-54-194-46-86.eu-west-1.compute.amazonaws.com.                                                  |
| elb-prod.unicornworld.local              | A     | Simple    | -         | dualstack.internal-internalelb-417588847.eu-west-1.elb.amazonaws.com.                              |
| enterprise-ad-dc1.unicornworld.local     | CNAME | Simple    | -         | ip-192-168-8-50.eu-west-1.compute.internal.                                                        |
| enterprise-ad-dc2.unicornworld.local     | CNAME | Simple    | -         | ip-192-168-9-55.eu-west-1.compute.internal.                                                        |
| enterprise-ad-ws1.unicornworld.local     | CNAME | Simple    | -         | ec2-54-154-166-132.eu-west-1.compute.amazonaws.com.                                                |
| enterprise-natinst.unicornworld.local    | CNAME | Simple    | -         | ec2-52-16-24-57.eu-west-1.compute.amazonaws.com.                                                   |
| enterprise-zc-connmgr.unicornworld.local | CNAME | Simple    | -         | ec2-52-17-11-65.eu-west-1.compute.amazonaws.com.                                                   |
| internalelbprod2.unicornworld.local      | CNAME | Simple    | -         | ip-192-168-9-104.eu-west-1.compute.internal.                                                       |
| internalelbtest1.unicornworld.local      | CNAME | Simple    | -         | ip-192-168-8-106.eu-west-1.compute.internal.                                                       |
| intranet.unicornworld.local              | A     | Simple    | -         | 192.168.4.30                                                                                       |
| ns1.unicornworld.local                   | A     | Simple    | -         | 172.31.3.5                                                                                         |
| ns.prod.unicornworld.local               | A     | Simple    | -         | 172.31.3.5                                                                                         |
| reinvent-labadmin.unicornworld.local     | CNAME | Simple    | -         | ec2-54-85-202-206.compute-1.amazonaws.com.                                                         |
| workstion.unicornworld.local             | CNAME | Simple    | -         | ec2-54-229-149-191.eu-west-1.compute.amazonaws.com.                                                |



Private unicornworld.local info

Delete zone Test record Configure query logging

▼ Hosted zone details

Edit hosted zone

|                                         |                             |                                                                                                                                |
|-----------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Hosted zone name<br>unicornworld.local  | Type<br>Private hosted zone | Associated VPCs<br>vpc-0b67c554f0a8cfa84   eu-west-1<br>vpc-00d13c704fac4a21e   eu-west-2<br>vpc-08da1c7a5f8005c8a   eu-west-2 |
| Hosted zone ID<br>Z0302818C3Z4A2PJ0Y8   | Record count<br>16          |                                                                                                                                |
| Description<br>Internal Unicorn Co Zone |                             |                                                                                                                                |



# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security



Putting it all  
together



# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security



Putting it all  
together

# Routing policies



# Routing policies



## Simple routing

Use if you want all of your clients to receive the same response(s).



## Weighted

Use when you have multiple resources that do the same job, and you want to specify the proportion of traffic that goes to each resource. For example: two or more EC2 instances.



## Geolocation

Use when you want to route traffic based on the location of your users.



## Latency

Use when you have resources in multiple AWS Regions and you want to route traffic to the Region that provides the best latency.



## Multivalue answer

Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random.



## Failover

Use to route traffic to a resource when the resource is healthy, or to a different resource when the first resource is unhealthy.

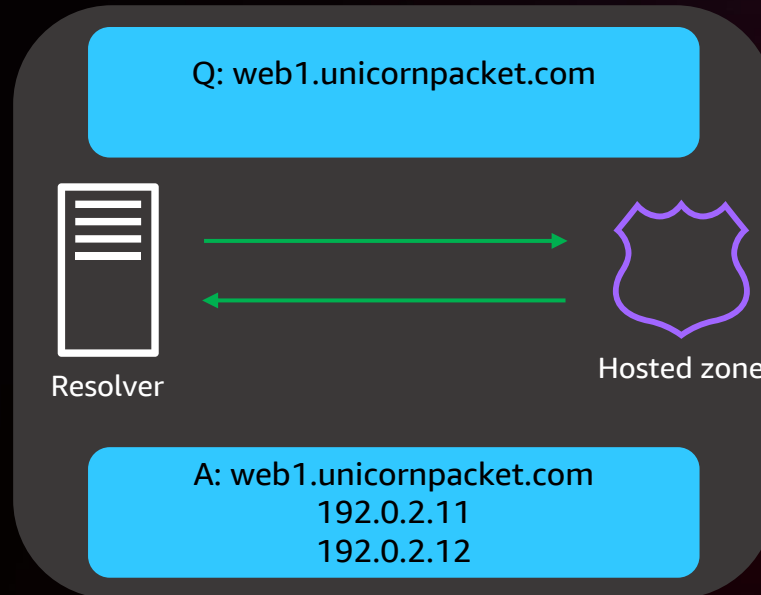


## IP-based

Use to route traffic to locations of IP address ranges in CIDR notation.

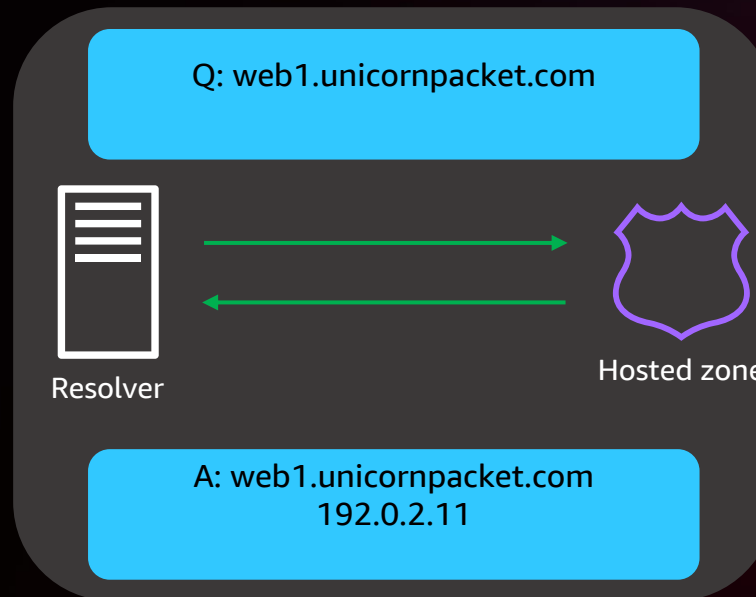


# Simple routing



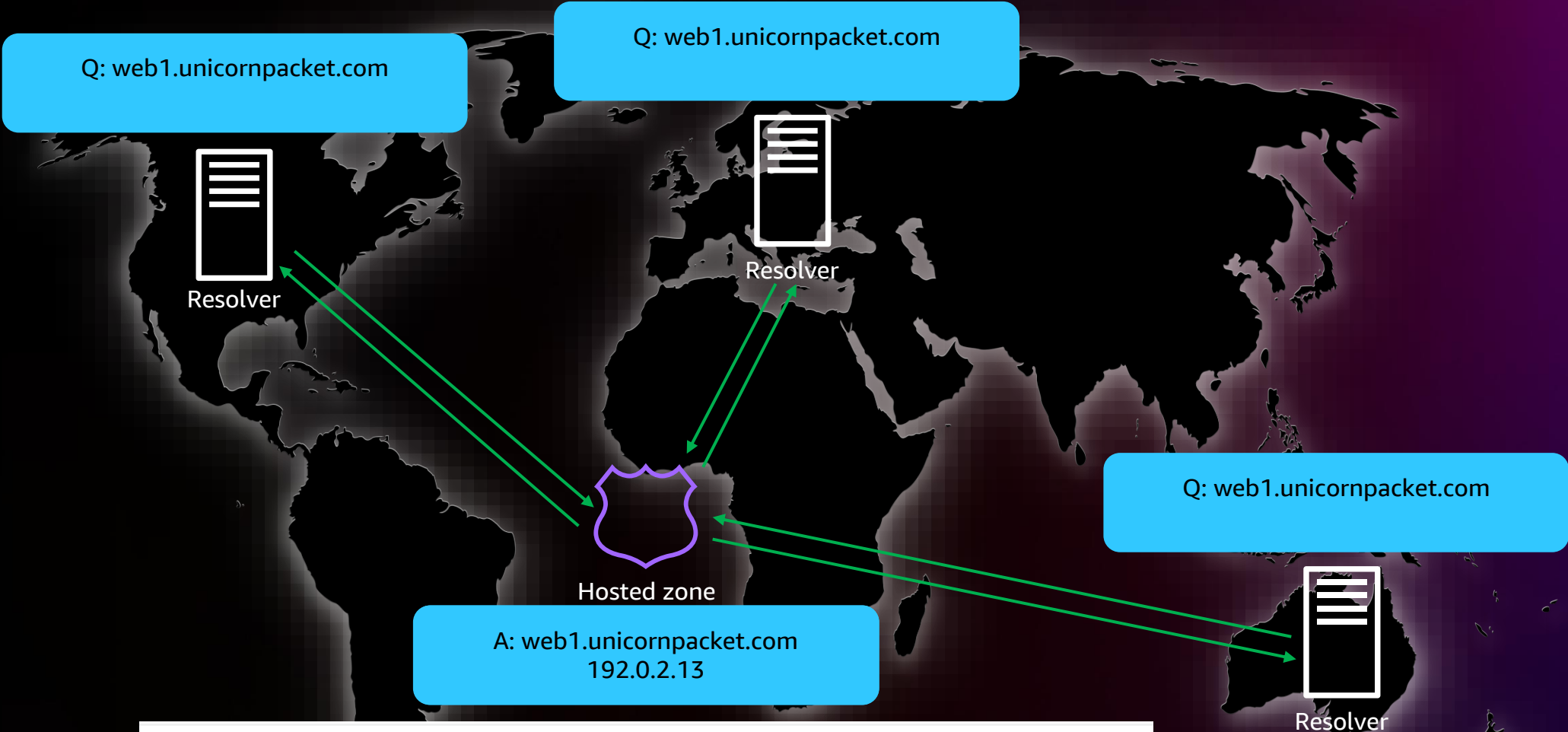
| <input type="checkbox"/> | Record name            | Type | Routing policy | Differentiator | Value/Route traffic to                                                                               |
|--------------------------|------------------------|------|----------------|----------------|------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | unicornpacket.com      | NS   | Simple         | -              | ns-254.awsdns-31.com.<br>ns-1762.awsdns-28.co.uk.<br>ns-1500.awsdns-59.org.<br>ns-563.awsdns-06.net. |
| <input type="checkbox"/> | unicornpacket.com      | SOA  | Simple         | -              | ns-254.awsdns-31.com. awsdns-hostmaster.amazon.com. 1<br>7200 900 1209600 86400                      |
| <input type="checkbox"/> | web1.unicornpacket.com | A    | Simple         | -              | 192.0.2.11<br>192.0.2.12                                                                             |

# Weighted



| <input type="checkbox"/> | Record name ▾          | Type ▾ | Routing policy ▾ | Differentiator ▾ | Value/Route traffic to ▾                                                                             |
|--------------------------|------------------------|--------|------------------|------------------|------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | unicornpacket.com      | NS     | Simple           | -                | ns-254.awsdns-31.com.<br>ns-1762.awsdns-28.co.uk.<br>ns-1500.awsdns-59.org.<br>ns-563.awsdns-06.net. |
| <input type="checkbox"/> | unicornpacket.com      | SOA    | Simple           | -                | ns-254.awsdns-31.com. awsdns-hostmaster.amazon.com. 1<br>7200 900 1209600 86400                      |
| <input type="checkbox"/> | web1.unicornpacket.com | A      | Weighted         | 50               | 192.0.2.11                                                                                           |
| <input type="checkbox"/> | web1.unicornpacket.com | A      | Weighted         | 30               | 192.0.2.12                                                                                           |

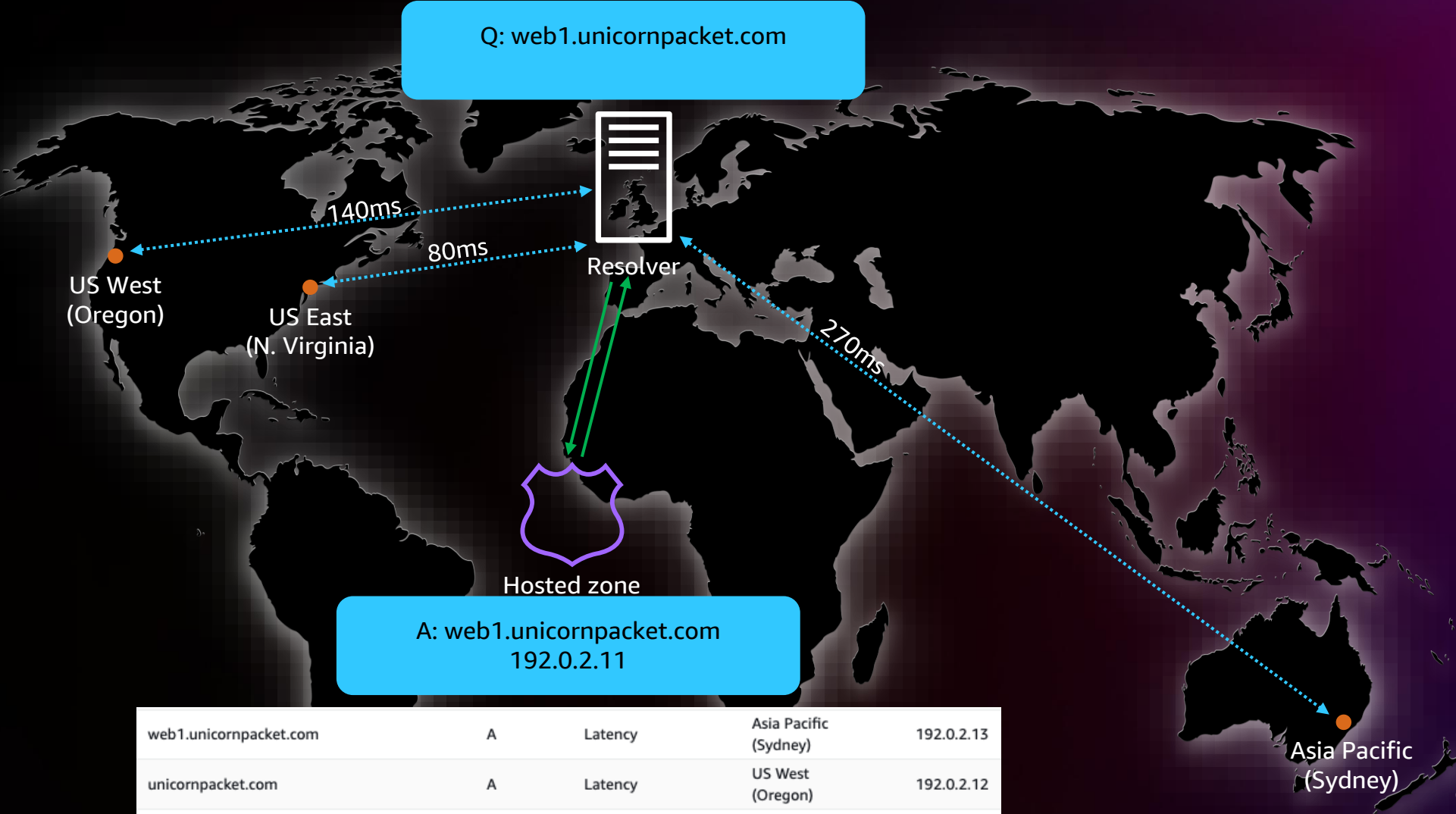
# Geolocation



|                        |   |             |               |            |
|------------------------|---|-------------|---------------|------------|
| web1.unicornpacket.com | A | Geolocation | North America | 192.0.2.11 |
| web1.unicornpacket.com | A | Geolocation | Europe        | 192.0.2.12 |
| web1.unicornpacket.com | A | Geolocation | Default       | 192.0.2.13 |

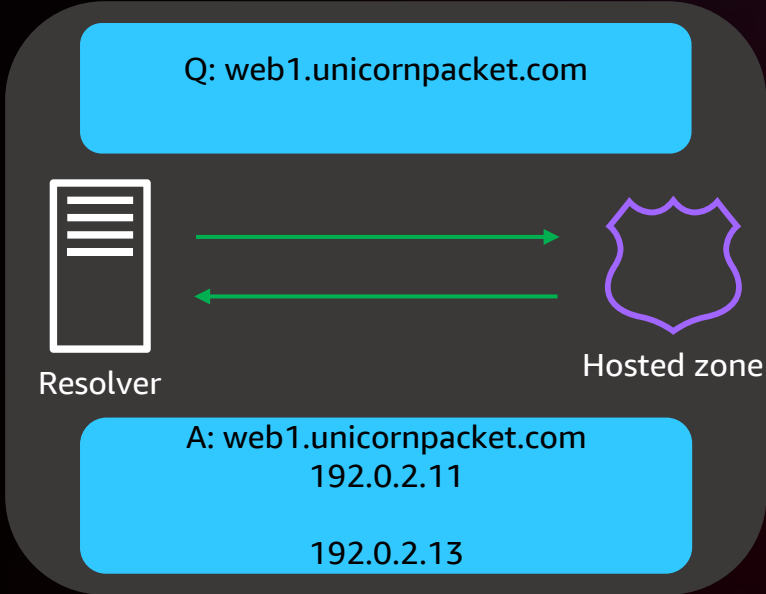


# Latency



|                        |   |         |                       |            |
|------------------------|---|---------|-----------------------|------------|
| web1.unicornpacket.com | A | Latency | Asia Pacific (Sydney) | 192.0.2.13 |
| unicornpacket.com      | A | Latency | US West (Oregon)      | 192.0.2.12 |
| web1.unicornpacket.com | A | Latency | US East (N. Virginia) | 192.0.2.11 |

# Multivalue answer



|                        |   |                                                                                                     |   |            |
|------------------------|---|-----------------------------------------------------------------------------------------------------|---|------------|
| web1.unicornpacket.com | A | Multivalue answer                                                                                   | - | 192.0.2.11 |
| web1.unicornpacket.com | A | <div><div></div><div>a day ago</div></div> <div><div></div><div>2 minutes ago</div></div> Unhealthy |   | 192.0.2.12 |
| web1.unicornpacket.com | A | Multivalue answer                                                                                   | - | 192.0.2.13 |



# IP-based routing



Q: web1.unicornpacket.com



Resolver/Client  
192.51.100.44



CIDR collections

Sydney: 203.0.113.0/24  
Default: 192.51.100.0/24



Hosted zone

A: web1.unicornpacket.com  
192.0.2.11

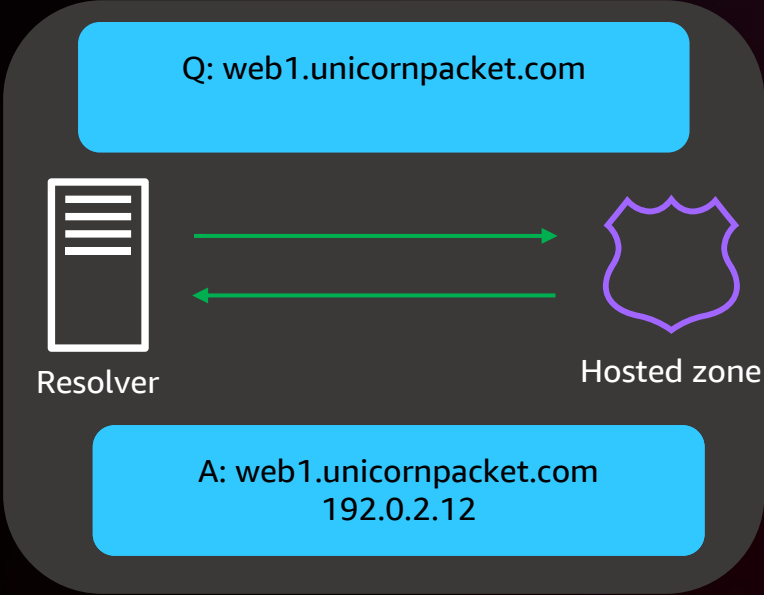
Q: web1.unicornpacket.com



Resolver/Client  
203.0.113.32

|                        |   |          |                           |            |
|------------------------|---|----------|---------------------------|------------|
| web1.unicornpacket.com | A | IP-based | Sydney <a href="#">↗</a>  | 192.0.2.11 |
| web1.unicornpacket.com | A | IP-based | Default <a href="#">↗</a> | 192.0.2.12 |

# Failover



|                        |                                                                    |         |          |           |            |
|------------------------|--------------------------------------------------------------------|---------|----------|-----------|------------|
| web1.unicornpacket.com | <div><div></div><div>a day ago</div><div>3 minutes ago</div></div> | Healthy | Failover | Primary   | 192.0.2.11 |
| web1.unicornpacket.com | A                                                                  |         | Failover | Secondary | 192.0.2.12 |

# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security



Putting it all  
together

# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security

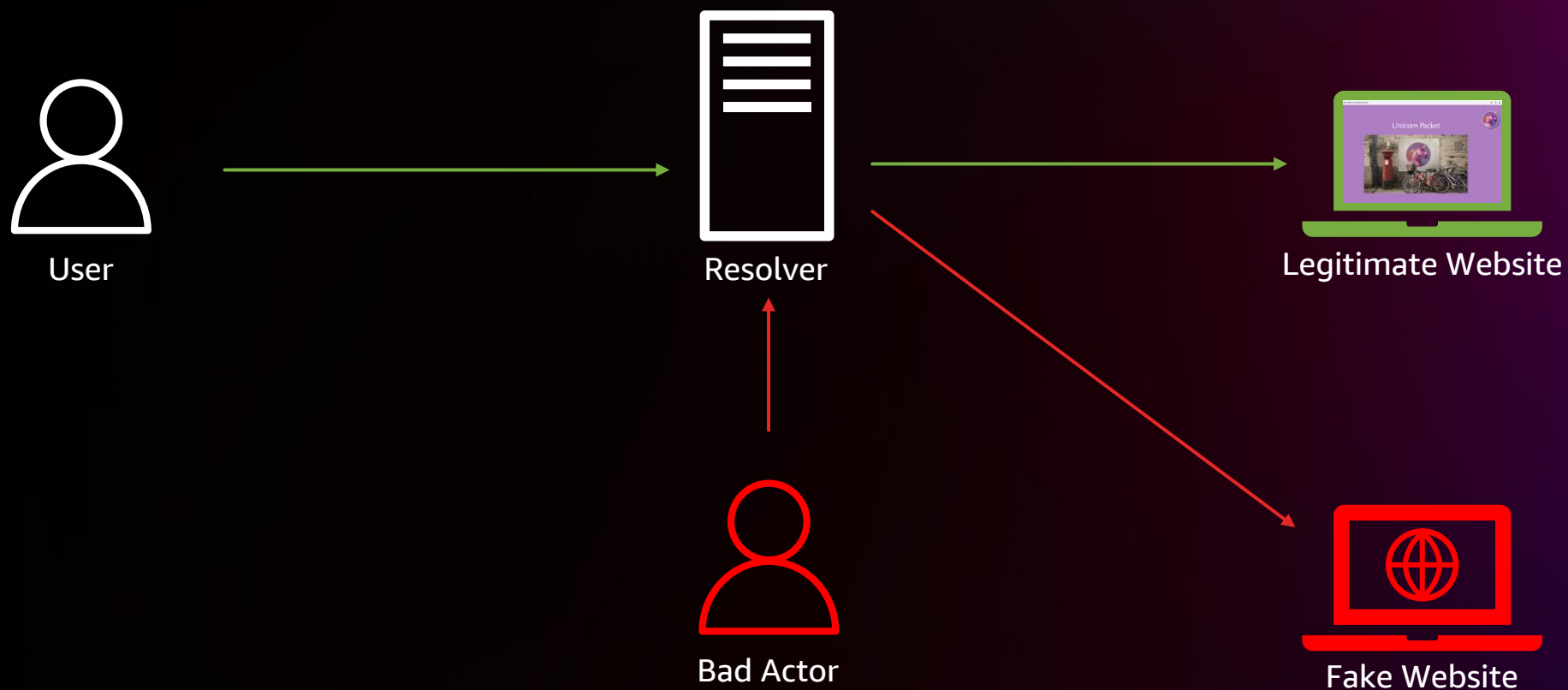


Putting it all  
together

# DNS security



# DNSSEC



# DNSSEC



Route 53 > Hosted zones > unicornpacket.com

Public unicornpacket.com Info

Delete zone Test record Configure query logging

► Hosted zone details Edit hosted zone

Records (4) **DNSSEC signing** Hosted zone tags (0)

**DNSSEC signing** Info

Enable DNSSEC signing

DNSSEC signing status

⊖ Not signing

**i You have not enabled DNSSEC signing for this hosted zone** X

To enable DNSSEC signing and have Route 53 create a key-signing key (KSK) for you, choose Enable DNSSEC signing. Next, you must establish a DNSSEC chain of trust for your hosted zone. You'll complete this step after you enable DNSSEC signing.

1. Enable DNSSEC signing

# DNSSEC



### Key-signing key (KSK) creation

On this page, Route 53 will create the key-signing key (KSK) for your hosted zone, based on a customer managed customer master key (CMK) that you choose.

Provide KSK name [Info](#)

Provide a name for the KSK that Route 53 creates for you automatically.

The name must have 3 - 128 characters. Valid characters: A-Z, a-z, and 0-9.

Customer managed CMK in AWS KMS [Info](#)

Route 53 creates the KSK for you based on a customer managed CMK in the AWS Key Management Service (AWS KMS). It's important that you don't change permissions or other configurations for the customer managed CMK after Route 53 uses it to create the KSK.

☐ Choose customer managed CMK

☒ Create customer managed CMK  
Additional charges apply.

Create customer managed CMK

Enter an alias for this key. Be aware that specific AWS KMS permissions are required to modify the key after it's created. [Learn more](#)

► Key properties

Cancel

Create KSK and enable signing

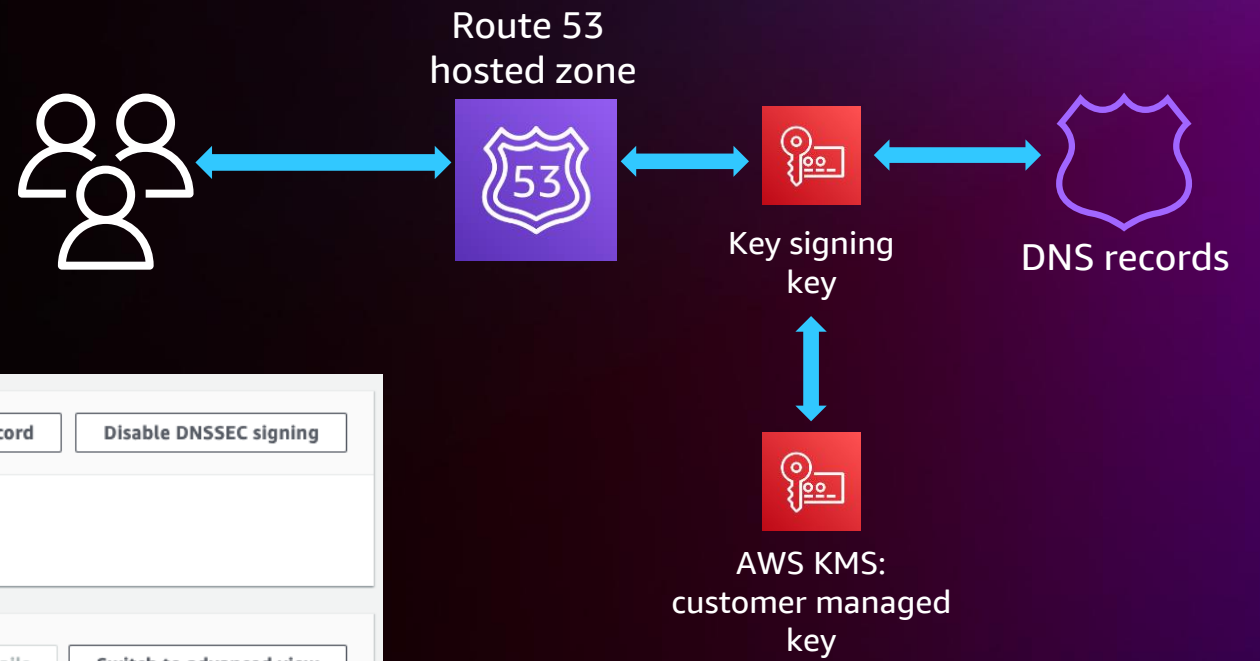


AWS KMS:  
customer managed  
key

## 2. Create a KSK



# DNSSEC



**DNSSEC signing** [Info](#)

[View information to create DS record](#) [Disable DNSSEC signing](#)

DNSSEC signing status  
✓ Signing

**Key-signing keys (KSKs)** [Info](#)

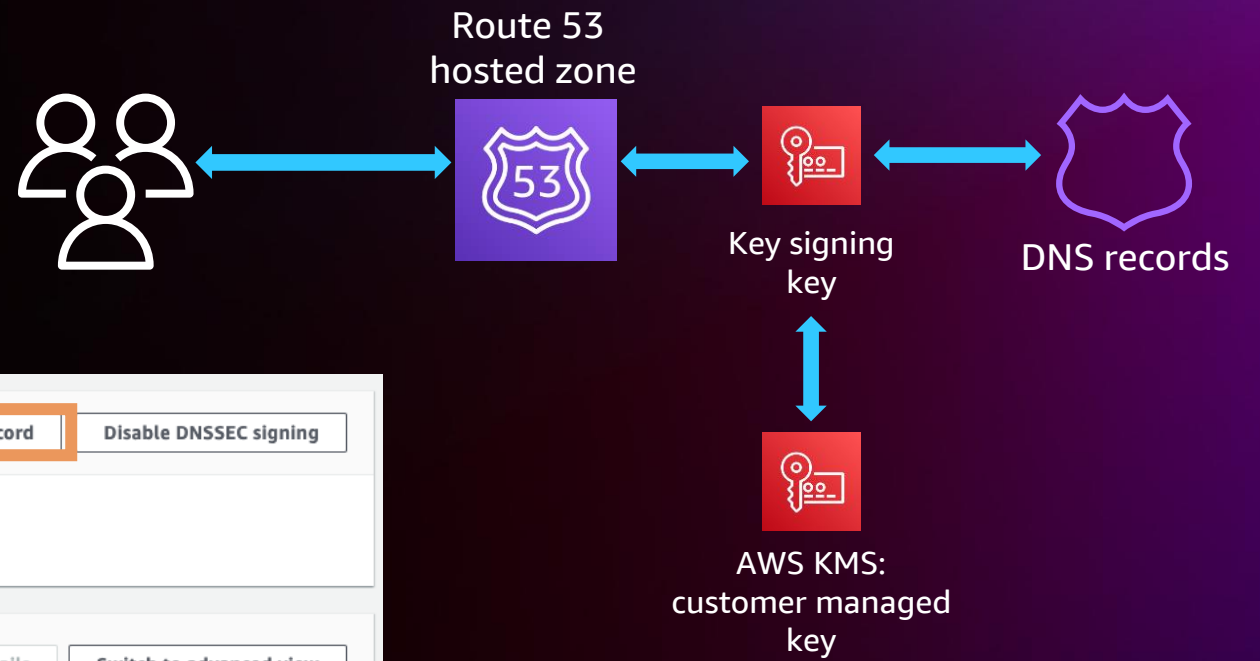
[View details](#) [Switch to advanced view](#)

< 1 > ⚙

|                       | Name             | Status                | Creation date                        |
|-----------------------|------------------|-----------------------|--------------------------------------|
| <input type="radio"/> | unicornpacketkey | <span>✓ Active</span> | October 11, 2022, 21:40 (UTC:+01:00) |

2. Create a KSK

# DNSSEC



**DNSSEC signing** [Info](#)

[View information to create DS record](#) [Disable DNSSEC signing](#)

DNSSEC signing status  
✔ Signing

**Key-signing keys (KSKs)** [Info](#)

[View details](#) [Switch to advanced view](#)

< 1 > ⚙

|                       | Name             | Status          | Creation date                        |
|-----------------------|------------------|-----------------|--------------------------------------|
| <input type="radio"/> | unicornpacketkey | ✔ <u>Active</u> | October 11, 2022, 21:40 (UTC:+01:00) |

3. Add DS record to parent

# DNSSEC



**▼ Establish a chain of trust** [Info](#)

To establish a chain of trust for DNSSEC, you must update the parent zone for your hosted zone with the DNSSEC information provided here. The updates that you make depend on if you use Route 53 or another registrar.

[▶ Route 53 registrar](#)

[▶ Another domain registrar](#)

**▼ Key-signing key (KSK) details**

|                              |                           |                                                       |                                         |
|------------------------------|---------------------------|-------------------------------------------------------|-----------------------------------------|
| KSK name<br>unicornpacketkey | Status<br>✔ <b>Active</b> | Creation date<br>October 11, 2022, 21:40 (UTC:+01:00) | Hosted zone ID<br>Z08033393PJJ58GDGJSFY |
|------------------------------|---------------------------|-------------------------------------------------------|-----------------------------------------|

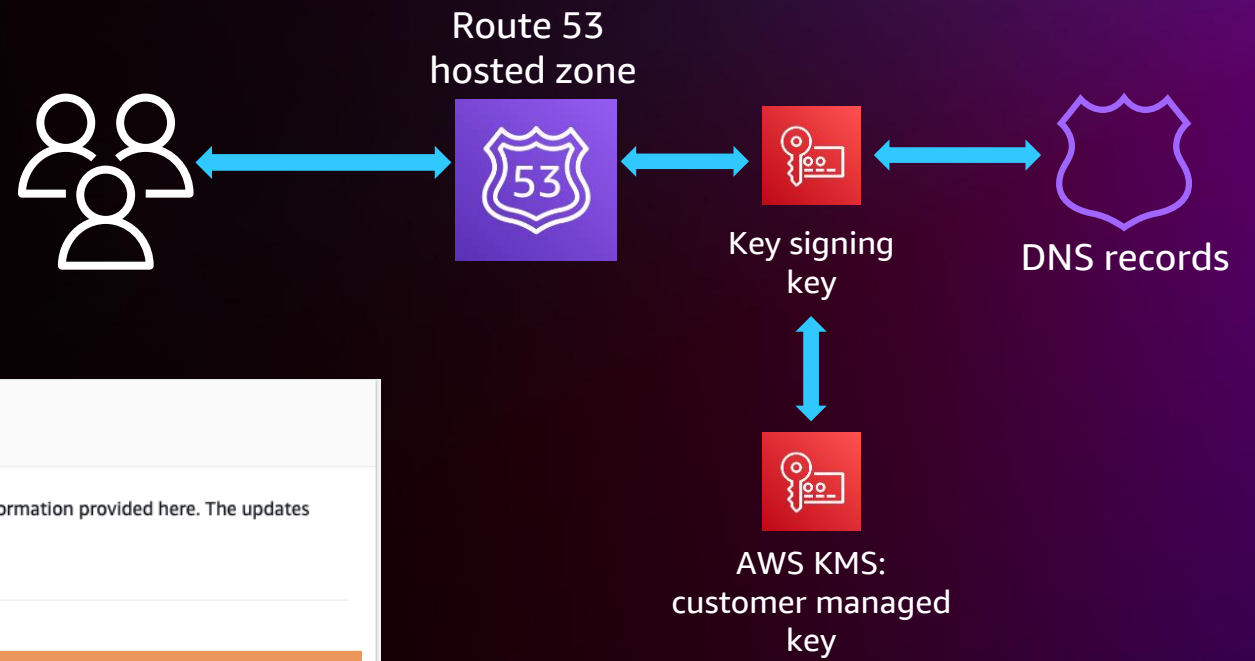
**▼ Customer managed CMK details** [Info](#) [Manage key](#)

The information for this customer master key (CMK) that you manage is from the Key Management System console.

|                                      |                                                                                         |                                                                                        |
|--------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Key alias<br>alias/unicornpacket-ksk | Status<br>✔ <b>Enabled</b><br><br>Creation date<br>October 11, 2022, 21:40 (UTC:+01:00) | Key ARN<br>arn:aws:kms:us-east-1:456989011791:key/5877a4a3-1cc9-4cfa-8dcc-972ea8caa50e |
|--------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

3. Add DS record to parent

# DNSSEC



## ▼ Establish a chain of trust [Info](#)

To establish a chain of trust for DNSSEC, you must update the parent zone for your hosted zone with the DNSSEC information provided here. The updates that you make depend on if you use Route 53 or another registrar.

### ▼ Route 53 registrar

Update your parent zone using the following information.

Key type

☒ KSK

Signing algorithm

☒ ECDSAP256SHA256

Public key

```
6dGaQHTkQLdFLkLu3HrVH0a4RTCI6eKVh
Q5XCWeO3O7Q8QpKHbrdT/LWjrKqLjedcDIRSM
J3ULAyYmtFX6HIdA==
```

► Another domain registrar

## 3. Add DS record to parent

# DNSSEC



## Registered domains > unicornpacket.com

Edit contacts

Manage DNS

Delete domain

|                     |                                       |                           |                                                                        |                 |                                                                                                                                              |
|---------------------|---------------------------------------|---------------------------|------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Domain              | unicornpacket.com                     | Transfer lock ⓘ           | Disabled (enable)                                                      | Name servers ⓘ  | ns-254.awsdns-31.com<br>ns-1762.awsdns-28.co.uk<br>ns-1500.awsdns-59.org<br>ns-563.awsdns-06.net<br><a href="#">Add or edit name servers</a> |
| Registration date ⓘ | 2022-10-04                            | Authorization code ⓘ      | <a href="#">Get code</a>                                               |                 |                                                                                                                                              |
| Expiration date ⓘ   | 2023-10-04 ( <a href="#">extend</a> ) | Domain name status code ⓘ | addPeriod<br>ok                                                        |                 |                                                                                                                                              |
| Auto renew ⓘ        | Enabled ( <a href="#">disable</a> )   | Tag ⓘ                     | View and manage tags for your domains using <a href="#">Tag editor</a> | DNSSEC status ⓘ | Disabled<br><a href="#">Manage keys</a>                                                                                                      |

3. Add DS record to parent

# DNSSEC

**i** After you configure DNSSEC with your DNS service, add the applicable public key to the domain. [Learn more](#)

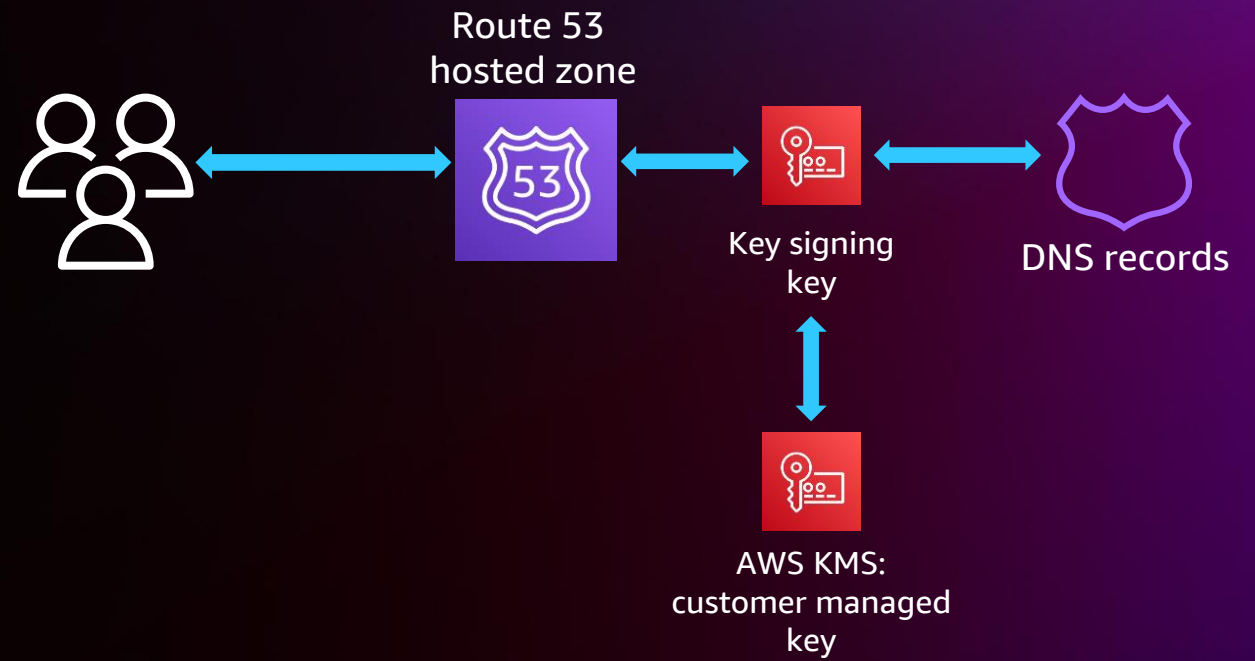
**✓** Your request for the creation of this DNSSEC entry was successfully submitted. It will be a few minutes before the listing is updated. You will receive an email when it is done, or if further action is necessary.

**Key type** 256 - ZSK ▾

**Algorithm** 13 - ECDSAP256SHA256 ▾

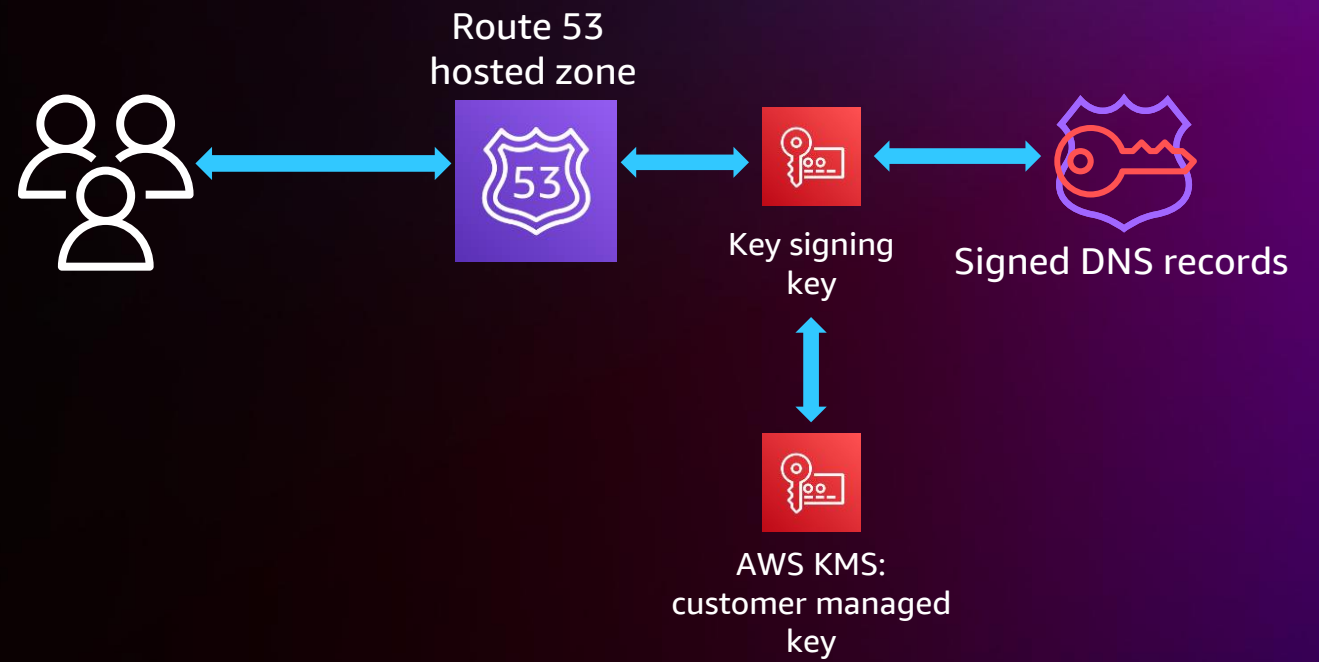
**Public key** 6dGaQHTkQLdFLkLu3HrVH0a4RTCl6eKVhQ5XCWeO3O7Q8QpKHbrdT/IWjrKqLjedcDIRS  
MJ3UIAyYmtFX6HIdA==

Add



3. Add DS record to parent

# DNSSEC



# DNSSEC

Route 53 > Resolver > VPCs

VPCs (2) [Info](#) 

| ID ▲                                  | Name ▼            | Rules ▼ | DNSSEC validation                                                                          |
|---------------------------------------|-------------------|---------|--------------------------------------------------------------------------------------------|
| <a href="#">vpc-00d13c704fac4a21e</a> | -                 | 1       |  Disabled |
| <a href="#">vpc-08da1c7a5f8005c8a</a> | unicornpacket-vpc | 1       |  Disabled |

4. Enable VPC DNSSEC validation



# DNSSEC

## DNSSEC validation

Use DNSSEC validation to check DNSSEC cryptographic signatures to ensure that a DNS response was not tampered with.

 Enabling DNSSEC validation can impact DNS resolution for public DNS records from AWS resources in a VPC, which could result in an outage. Be aware that enabling or disabling DNSSEC validation can take several minutes.

☒ Enable DNSSEC validation on this VPC

Status

 Enabling

4. Enable VPC DNSSEC validation



# DNSSEC

Route 53 > Resolver > VPCs

VPCs (2) [Info](#) 

| ID ▲                                  | Name ▼            | Rules ▼ | DNSSEC validation                                                                          |
|---------------------------------------|-------------------|---------|--------------------------------------------------------------------------------------------|
| <a href="#">vpc-00d13c704fac4a21e</a> | -                 | 1       |  Disabled |
| <a href="#">vpc-08da1c7a5f8005c8a</a> | unicornpacket-vpc | 1       |  Enabled  |

4. Enable VPC DNSSEC validation

# DNSSEC

```
[ec2-user@ip-10-0-131-49 ~]$ clear
[ec2-user@ip-10-0-131-49 ~]$ dig unicornpacket.com +trace
; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.amzn2.5.2 <<>> unicornpacket.com +trace
;; global options: +cmd
518400 IN NS K.ROOT-SERVERS.NET.
518400 IN NS L.ROOT-SERVERS.NET.
518400 IN NS M.ROOT-SERVERS.NET.
518400 IN NS A.ROOT-SERVERS.NET.
518400 IN NS B.ROOT-SERVERS.NET.
518400 IN NS C.ROOT-SERVERS.NET.
518400 IN NS D.ROOT-SERVERS.NET.
518400 IN NS E.ROOT-SERVERS.NET.
518400 IN NS F.ROOT-SERVERS.NET.
518400 IN NS G.ROOT-SERVERS.NET.
518400 IN NS H.ROOT-SERVERS.NET.
518400 IN NS I.ROOT-SERVERS.NET.
518400 IN NS J.ROOT-SERVERS.NET.
;; Received 239 bytes from 10.0.0.2#53(10.0.0.2) in 0 ms
```

```
com. 172800 IN NS e.gtld-servers.net.
com. 172800 IN NS b.gtld-servers.net.
com. 172800 IN NS a.gtld-servers.net.
com. 172800 IN NS d.gtld-servers.net.
com. 172800 IN NS i.gtld-servers.net.
com. 172800 IN NS f.gtld-servers.net.
com. 172800 IN NS j.gtld-servers.net.
com. 172800 IN NS k.gtld-servers.net.
com. 172800 IN NS c.gtld-servers.net.
com. 172800 IN NS g.gtld-servers.net.
com. 172800 IN NS h.gtld-servers.net.
com. 172800 IN NS l.gtld-servers.net.
com. 172800 IN NS m.gtld-servers.net.
```

```
com. 86400 IN RRSIG DS 8 1 86400 20221125050000 2022112040000 18733 . I5it4epK25laf74ZFhYWR/3w
GjsFroYCirdzIyMrzs4vJMU0XLVs+Ws veV+vnXqmVuN5HkJJwCR5Ff43mCZ+KhypfArYQ/RNtGazBKz0gX5wxZ6 2ub15Vr3318yAMt13iN2N44RB6ZBiEKluri
heBFoBFqW8LpR7/89PqYV 1H0TKZGDKGCNZedLPa+ukQcgp2YRdXOZFV3kNgXeKGlXyFk9Nv8eVqnP Tc5bQY1D1ZvU026MB95JG4LY0v4AXPt0gu0iyenC50RT
dVbiq/R2PHj hn92tvVz0ZkmyWvchHTP5uuyRAM/ktkqQL+R6i9B+/fqIdyuEw2YEgZI ktN0hw==
;; Received 1177 bytes from 192.5.5.241#53(F.ROOT-SERVERS.NET) in 2 ms
```

```
unicornpacket.com. 172800 IN NS ns-254.awsdns-31.com.
unicornpacket.com. 172800 IN NS ns-563.awsdns-06.net.
unicornpacket.com. 172800 IN NS ns-1762.awsdns-28.co.uk.
unicornpacket.com. 172800 IN NS ns-1500.awsdns-59.org.
```

```
unicornpacket.com. 86400 IN RRSIG DS 8 2 86400 20221116072107 20221109061107 53929 com. I3giP8YA5WfVtWbJpgU8Ql
9gTW2ou0Ic7KiWt6MnR4MTXGZv/PnJAiz/ p2R6gh+MJAXlC9Fg29F0r+9nQ0lv0dV/4rS5Ieo0kzQ3tP7YZVyTalfc ee0jzq+70hUYEkXjlyjvGH/vFaImJwI.
jF5YI1QFv/RBc4v3m7BZc2Yy 65YyWNLkSAGySoGLneMCGdyX7plq4ymqF3/Aupu1T5Tx7g==
```

```
unicornpacket.com. 900 IN RRSIG SOA 13 2 900 2022112165554 2022112144054 25012 unicornpacket.com. LW1GM5s
v+zTziXy/S05L09c05FWCNuoCPn0dDs0oESaHEIG30Nooypr j/CiVBGHwBmAvyU3h4wcJwHKbo25pw==
unicornpacket.com. 86400 IN NSEC \000.unicornpacket.com. NS SOA PTR HINFO MX TXT RP AAAA SRV NAPTR DNAME RRS
G NSEC DNSKEY NSEC3PARAM SPF IXFR AXFR CAA
unicornpacket.com. 86400 IN RRSIG NSEC 13 2 86400 20221113164054 2022112144054 25012 unicornpacket.com. tCm6
VzQR+0WKRckYB2+g7hXrU5MT4shWbowMmPQwFRnMNpkPcu/lBp 9XVPmNTM9/LZLNCzbG72zmlcYXMDbQ==
;; Received 420 bytes from 205.251.192.254#53(ns-254.awsdns-31.com) in 9 ms
```

```
[ec2-user@ip-10-0-131-49 ~]$ dig
; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.amzn2.5.2 <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 53128
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
; . IN A

;; Query time: 8 msec
;; SERVER: 10.0.0.2#53(10.0.0.2)
;; WHEN: Sat Nov 12 15:46:16 UTC 2022
;; MSG SIZE rcvd: 46
```



```
dig
; <<>> DiG 9.10.6 <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 37208
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
; . IN A

;; ANSWER SECTION:
. 281 IN A

;; Query time: 27 msec
;; SERVER:
;; WHEN: Sun Oct 16 10:33:07 BST 2022
;; MSG SIZE rcvd: 62
```



# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security



Putting it all  
together

# Agenda



What is a  
domain name?



What is  
Route 53?



Routing  
policies



DNS  
security



Putting it all  
together

# Transferring a domain

# Transfers – things to note

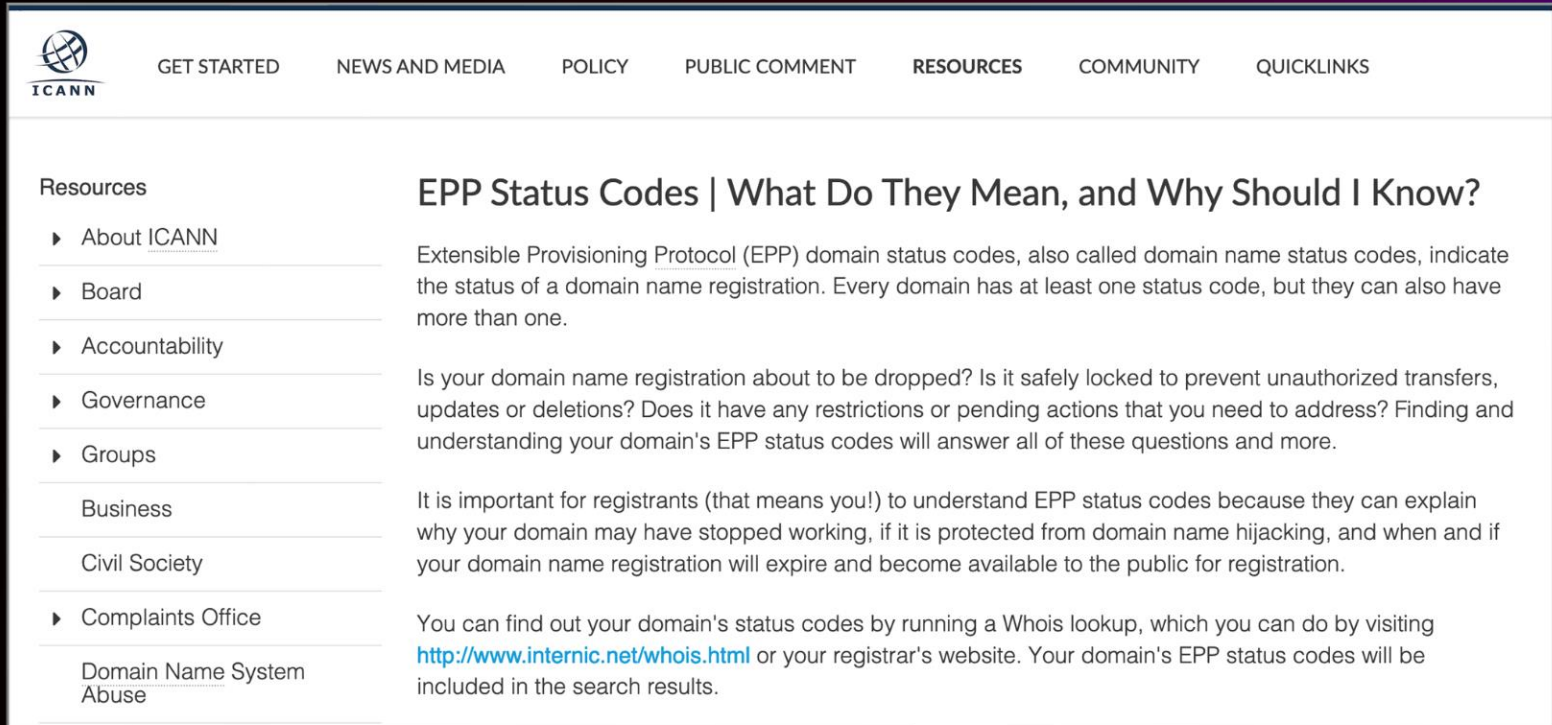
- Does AWS support the TLD
  - Dates – registration & expiry
  - Quotas – default per account is 20
  - Nameservers – max 6 on Route 53
- 
- Charges & support
  - Test first
  - Repeat in bulk



# Domain status

For TLDs – cannot transfer if status =

- clientTransferProhibited
- pendingDelete
- pendingTransfer
- redemptionPeriod
- serverTransferProhibited



The screenshot shows the ICANN website's 'Resources' section. The header includes the ICANN logo and navigation links: GET STARTED, NEWS AND MEDIA, POLICY, PUBLIC COMMENT, RESOURCES, COMMUNITY, and QUICKLINKS. The 'Resources' sidebar lists: About ICANN, Board, Accountability, Governance, Groups, Business, Civil Society, Complaints Office, and Domain Name System Abuse. The main content area is titled 'EPP Status Codes | What Do They Mean, and Why Should I Know?'. It explains that Extensible Provisioning Protocol (EPP) domain status codes, also called domain name status codes, indicate the status of a domain name registration. Every domain has at least one status code, but they can also have more than one. It then asks if the domain name registration is about to be dropped, if it is safely locked to prevent unauthorized transfers, updates or deletions, and if it has any restrictions or pending actions. Finding and understanding the domain's EPP status codes will answer all these questions. It emphasizes that it is important for registrants to understand EPP status codes because they can explain why a domain may have stopped working, if it is protected from domain name hijacking, and when and if the domain name registration will expire and become available to the public. It concludes by stating that you can find out your domain's status codes by running a Whois lookup, which can be done by visiting <http://www.internic.net/whois.html> or your registrar's website. Your domain's EPP status codes will be included in the search results.



# High-level steps

- Test the process and all the steps
- Separate the tasks – delegation & registrar transfer
- Drop the TTL & disable DNSSEC
- Create your public hosted zone in Route 53
- Note the delegation set of nameservers
- Update existing registrar delegation to point to Route 53 nameservers
- Transfer the domain to the Route 53 registrar
- [re-]Enable DNSSEC

# Create your public hosted zone in Route 53

- Create your public hosted zone in Route 53
  - Note the delegation set provided – the authoritative nameservers for the zone
- Prepare the zone contents in advance
  - Start by matching the contents of the current zone
    - Export from current provider – “standard BIND format”
    - Import – Route 53 “import zone file” or cli53 (third-party tool)
- Improve the contents to use AWS specific features after migration

# Zone file import process

```
$ORIGIN unicornpacket.com.
@ 900 IN SOA ns1.unicorndns.net.
 2022021401
 7200
 900
 1209600
 86400

@ 600 IN MX 10 mail.unicornpacket.com.
@ 172800 IN NS ns1.unicorndns.net.
@ 172800 IN NS ns2.unicorndns.net.
@ 172800 IN NS ns3.unicorndns.net.
@ 172800 IN NS ns4.unicorndns.net.

mail 600 IN A 203.0.113.21
mail 600 IN AAAA 2001:db8::113:21

www 600 IN A 198.51.100.99
www 600 IN AAAA 2001:db8::100:99

portal 600 IN CNAME portal.colourfulportals.com.
```

# Zone file import process

The screenshot displays the AWS Route 53 console interface. The top navigation bar shows the path: Route 53 > Hosted zones > unicornpacket.com. The main header for the hosted zone 'unicornpacket.com' includes buttons for 'Delete zone', 'Test record', 'Configure query logging', and 'Edit hosted zone'. Below this, the 'Hosted zone details' section is visible.

The 'Import zone file' modal is open, showing the 'Import zone file' page. It includes a description: 'You can create records for a Route 53 hosted zone by importing a zone file.' The 'Zone file' section instructs the user to 'Paste the contents of your zone file below.' The zone file content is as follows:

```
$ORIGIN unicornpacket.com.
@ 900 IN SOA ns1.unicorndns.net.
 hostmaster.unicorndns.net.
 2022021401
 7200
 900
 1209600
 86400

@ 600 IN MX 10 mail.unicornpacket.com.
@ 172800 IN NS ns1.unicorndns.net.
@ 172800 IN NS ns2.unicorndns.net.
@ 172800 IN NS ns3.unicorndns.net.
@ 172800 IN NS ns4.unicorndns.net.

mail 600 IN A 203.0.113.21
mail 600 IN AAAA 2001:db8::113:21

www 600 IN A 198.51.100.99
www 600 IN AAAA 2001:db8::100:99

portal 600 IN CNAME portal.colourfulportals.com.
```

Below the zone file content, a note states: 'If the hosted zone already contains records that appear in the zone file, the import process fails, and no records are created. Enter multiple records on separate lines.'

In the background, the 'Records' section of the hosted zone is visible. It includes a table with columns 'Routin...', 'Differ...', and 'Value/Route traffic to'. The table lists several records, including 'Simple' records pointing to 'ns-1318.awsdns-36.org' and 'ns-238.awsdns-29.com'. The 'Import zone file' button in the top right of the records section is circled in red.

# Zone file import process

```
$ORIGIN unicornpacket.com.
@ 900 IN SOA ns1.unicorndns.net
hostmaster.unicorndns.net.
2022021401
7200
900
1209600
86400

@ 600 IN MX 10 mail.unicornpacket.com.
@ 172800 IN NS ns1.unicorndns.net.
@ 172800 IN NS ns2.unicorndns.net.
@ 172800 IN NS ns3.unicorndns.net.
@ 172800 IN NS ns4.unicorndns.net.

mail 600 IN A 203.0.113.21
mail 600 IN AAAA 2001:db8::113:21

www 600
www 600

portal 600 IN
```

### Record preview for unicornpacket.com (6)

Route 53 creates the following records when you choose Import zone file. If you edit the contents of the zone file above, the table reflects your changes.

< 1 >

⚙

| Record name ▲             | Type ▼ | Value/Route traffic to ▼     | TTL (seconds) ▼ |
|---------------------------|--------|------------------------------|-----------------|
| mail.unicornpacket.com.   | A      | 203.0.113.21                 | 600             |
| mail.unicornpacket.com.   | AAAA   | 2001:db8::113:21             | 600             |
| portal.unicornpacket.com. | CNAME  | portal.colourfulportals.com. | 600             |
| unicornpacket.com.        | MX     | 10 mail.unicornpacket.com.   | 600             |
| www.unicornpacket.com.    | A      | 198.51.100.99                | 600             |
| www.unicornpacket.com.    | AAAA   | 2001:db8::100:99             | 600             |



# Update the delegation

At your existing provider/registrar

Modify the nameservers

Provide the Route 53 delegation set

Be patient ... time to live (TTL) needs to expire

Test using a new record only present in Route 53



# Remember earlier ... now is the time

- Register a new domain name
- Transfer an existing supported domain name

1: Select Domain

2: Domain Options

3: Contact Details

4: Review & Purchase

## Transfer Domain to Route 53

You can transfer registration for one or more domains from another registrar to Route 53. Before you continue, do the following:

- Confirm that the domain is transferable. See [Transfer requirements for top-level domains](#).
- For each domain that you want to transfer, perform the first four steps of [Transferring registration for a domain to Route 53](#).

To transfer up to five domains, you can enter each domain name below.  
To transfer more than five domains, you can use the [Transfer multiple domains to Route 53 page](#).

Check

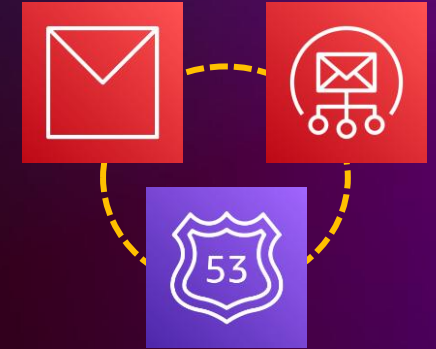
unicornpacket.com can be transferred to Route 53 (\$12.00)

To transfer the domain to Route 53 please ensure the following:

- Disable domain privacy for the domain so we can get your contact information.
- Ensure the domain's registrant email at WHOIS is valid and reachable.
- Get an authorization code for the domain (*Not required for certain TLDs*).

Add to Cart

# Implement other AWS Services



Amazon WorkMail > Organizations > unicornpacket > Domains > unicornpacket.com

unicornpacket.com ↻ Copy all Update all in Route 53

To be able to use this domain with WorkMail, you need to configure it by adding the following DNS records. Since your domain is hosted in Route 53, we can configure it automatically by pressing "Update all in Route 53" button above. Alternatively, you can copy-paste the below DNS records one by one or use "Copy all" button above. Note that after you make any changes, it can take up to 72 hours before all DNS records are propagated.

**Domain ownership** [Info](#)

▼ Details

| Type | Record name                                   | Value                                                 | Status                      |
|------|-----------------------------------------------|-------------------------------------------------------|-----------------------------|
| TXT  | <a href="#">_amazonses.unicornpacket.com.</a> | <a href="#">F7oPICFhcbHu4lRcm/xjA0bnC4dBS6V7nM...</a> | <span>⚠ Inconsistent</span> |

**WorkMail configuration** [Info](#)

▼ Details

| Type  | Record name                                     | Value                                                    | Status                 |
|-------|-------------------------------------------------|----------------------------------------------------------|------------------------|
| MX    | <a href="#">unicornpacket.com.</a>              | <a href="#">10 inbound-smtp.eu-west-1.amazonaws.c...</a> | <span>⚠ Missing</span> |
| CNAME | <a href="#">autodiscover.unicornpacket.com.</a> | <a href="#">autodiscover.mail.eu-west-1.awsapps.com.</a> | <span>⚠ Missing</span> |

"Easy" button

**Custom MAIL FROM domain** [Info](#)

Messages sent through Amazon SES will be marked as originating from your domain instead of a subdomain

Publish DNS records to Route53 Edit

|                                 |                        |
|---------------------------------|------------------------|
| MAIL FROM configuration         | MAIL FROM domain       |
| <span>⚠ Temporary failed</span> | mail.unicornpacket.com |

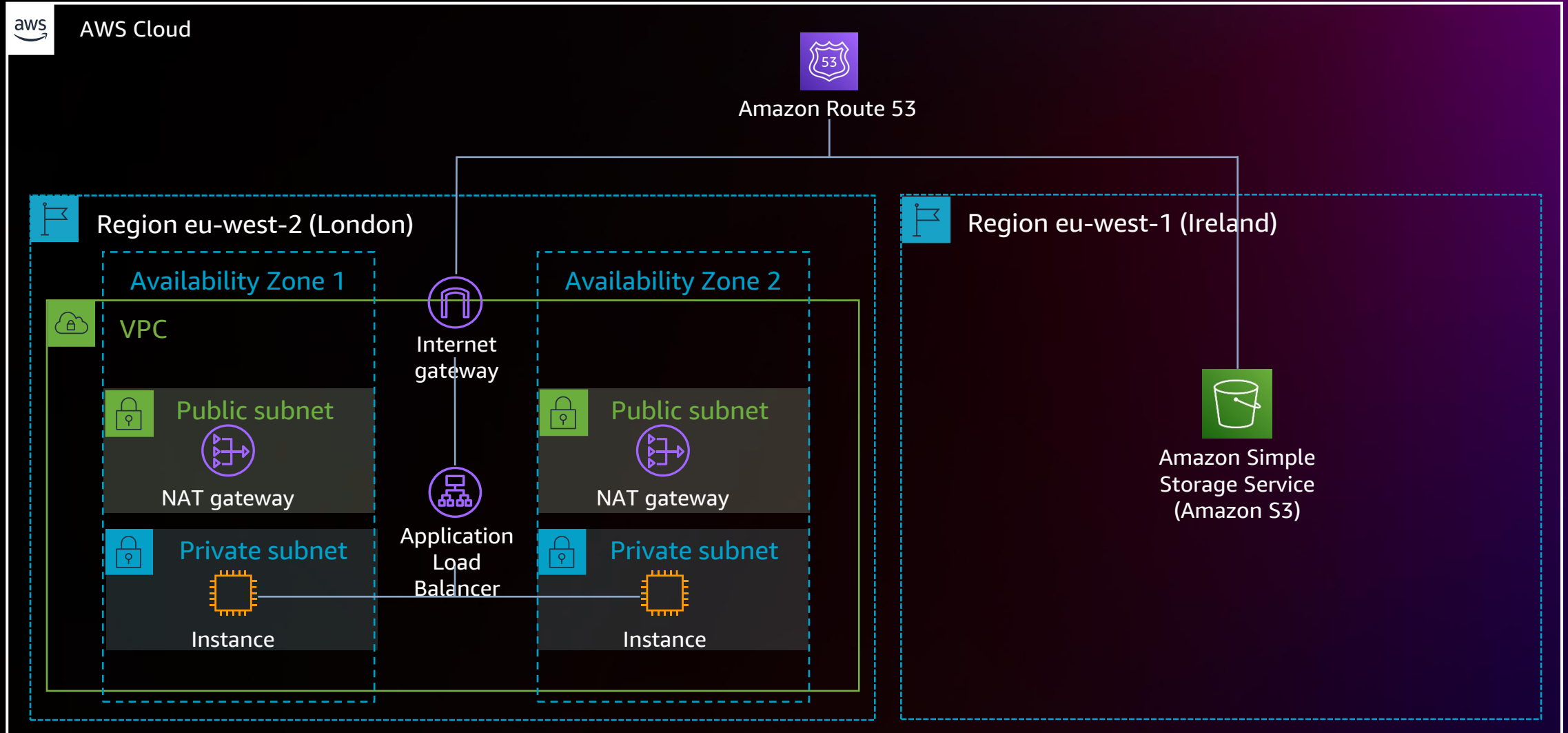


# Update the public hosted zone

| Records (14) <a href="#">Info</a>                                                                                                |                                                               |        |             |             |                                                                                                      | <a href="#">Delete record</a> | <a href="#">Import zone file</a> | <a href="#">Create record</a> |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------|-------------|-------------|------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------|-------------------------------|
| Automatic mode is the current search behavior optimized for best filter results. <a href="#">To change modes go to settings.</a> |                                                               |        |             |             |                                                                                                      |                               |                                  |                               |
| <input type="text" value="Filter records by property or value"/>                                                                 |                                                               |        |             |             |                                                                                                      | Type ▼                        | Routing policy ▼                 | Alias ▼                       |
|                                                                                                                                  |                                                               |        |             |             |                                                                                                      | < 1 > ⚙                       |                                  |                               |
| <input type="checkbox"/>                                                                                                         | Record name ▼                                                 | Type ▼ | Routin... ▼ | Differ... ▼ | Value/Route traffic to ▼                                                                             |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | unicornpacket.com                                             | MX     | Simple      | -           | 10 inbound-smtp.eu-west-1.amazonaws.com.                                                             |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | unicornpacket.com                                             | NS     | Simple      | -           | ns-254.awsdns-31.com.<br>ns-1762.awsdns-28.co.uk.<br>ns-1500.awsdns-59.org.<br>ns-563.awsdns-06.net. |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | unicornpacket.com                                             | SOA    | Simple      | -           | ns-254.awsdns-31.com. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400                         |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | unicornpacket.com                                             | TXT    | Simple      | -           | "v=spf1 include:amazonses.com ~all"                                                                  |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | _amazonses.unicornpacket.com                                  | TXT    | Simple      | -           | "F7oPICFhcbHu4lRcm/xjA0bnC4dBS6V7nMRsB8LXQeQ="                                                       |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | _dmarc.unicornpacket.com                                      | TXT    | Simple      | -           | "v=DMARC1;p=quarantine;pct=100;fo=1"                                                                 |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | gorzhgnkcc72apewqfggoyh2acmjympu._domainkey.unicornpacket.com | CNAME  | Simple      | -           | gorzhgnkcc72apewqfggoyh2acmjympu.dkim.amazonses.com.                                                 |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | mlsl7utli6xuci4ikvbdjknkctgsl3eq._domainkey.unicornpacket.com | CNAME  | Simple      | -           | mlsl7utli6xuci4ikvbdjknkctgsl3eq.dkim.amazonses.com.                                                 |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | tklt5hzx7ouhhume7uv63tkba3xbqrub._domainkey.unicornpacket.com | CNAME  | Simple      | -           | tklt5hzx7ouhhume7uv63tkba3xbqrub.dkim.amazonses.com.                                                 |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | autodiscover.unicornpacket.com                                | CNAME  | Simple      | -           | autodiscover.mail.eu-west-1.awsapps.com.                                                             |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | mail.unicornpacket.com                                        | MX     | Simple      | -           | 10 feedback-smtp.eu-west-1.amazonses.com                                                             |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | mail.unicornpacket.com                                        | TXT    | Simple      | -           | "v=spf1 include:amazonses.com ~all"                                                                  |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | www.web1.unicornpacket.com                                    | A      | Failover    | Primary     | dualstack.lb-unicorn-packets-52827961.eu-west-2.elb.amazonaws.com.                                   |                               |                                  |                               |
| <input type="checkbox"/>                                                                                                         | www.web1.unicornpacket.com                                    | A      | Failover    | Secondary   | s3-website-eu-west-1.amazonaws.com.                                                                  |                               |                                  |                               |

# Configuring a routing policy

# Let's set up failover



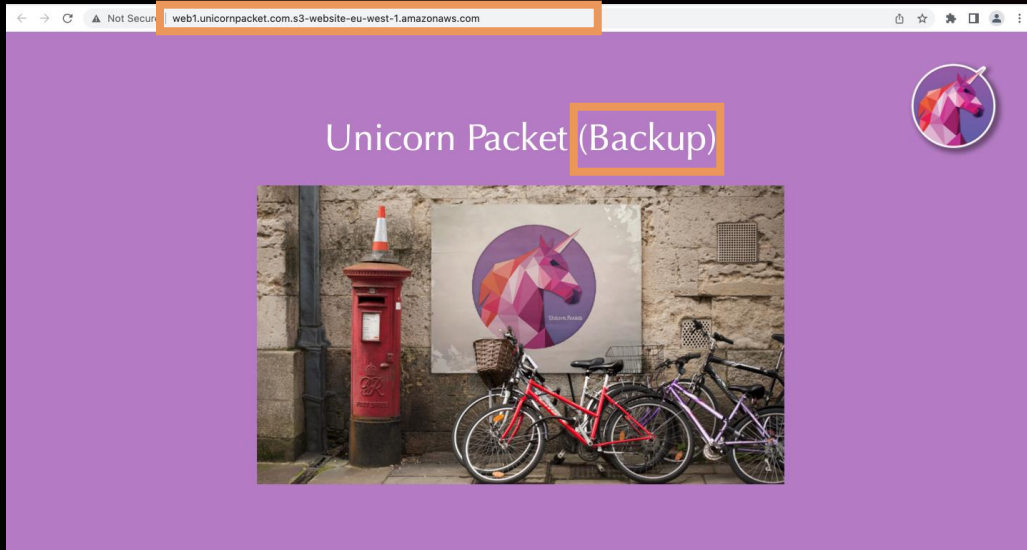
# Website hosting



Amazon Simple Storage  
Service (Amazon S3)



Application Load  
Balancer



## Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

Enabled

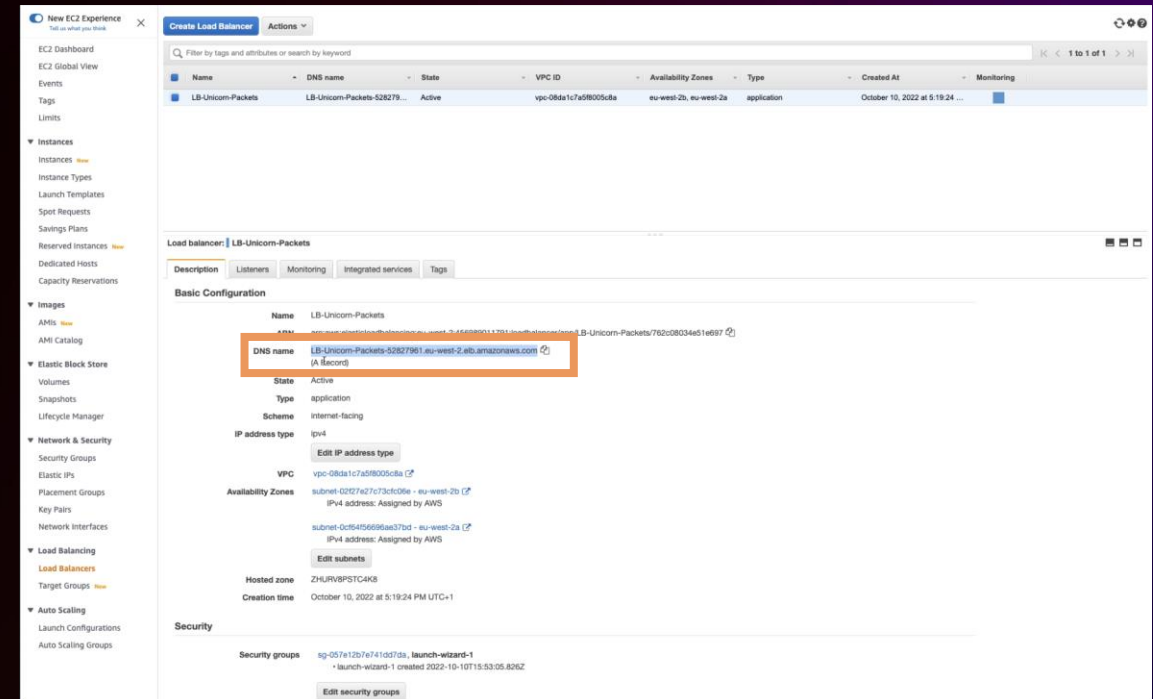
Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://www.web1.unicornpacket.com.s3-website-eu-west-1.amazonaws.com>



# Health checks

Configure two health checks to monitor the Application Load Balancer and the Amazon S3 endpoint

Dashboard

Hosted zones

Health checks

IP-based routing

CIDR collections

Traffic flow

Traffic policies

Policy records

Domains

Registered domains

Pending requests

Resolver

VPCs

Inbound endpoints

Outbound endpoints

Rules

Query logging

DNS Firewall

Rule groups

Domain lists

Application Recovery Controller

Getting started

Readiness check

Resource sets

Readiness rules

Routing control

Clusters

Health check with id 99e05ac7-075b-4106-926e-09588f84ea1c has been created successfully

Create health check

Delete health check

Edit health check

Filter by keyword

<< 1 to 2 of 2 health checks >>

| Name                                     | Status  | Description                              | Alarms                | ID                                   |
|------------------------------------------|---------|------------------------------------------|-----------------------|--------------------------------------|
| <input checked="" type="checkbox"/> web1 | Healthy | http://LB-Unicorn-Packets-52827961.eu... | No alarms configured. | 5091d7db-df84-4664-bf47-45948e13e3b1 |
| <input type="checkbox"/> web2            | Healthy | http://www.web1.unicompacket.com.s3...   | No alarms configured. | 99e05ac7-075b-4106-926e-09588f84ea1c |

Info

Monitoring

Alarms

Tags

Health checkers

Latency

View current status

View last failed check

Refresh

| Health checker region     | Health checker IP | Last checked                | Status                                                         |
|---------------------------|-------------------|-----------------------------|----------------------------------------------------------------|
| Asia Pacific (Tokyo)      | 15.177.42.70      | Oct 11, 2022 9:49:36 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| Asia Pacific (Tokyo)      | 15.177.46.70      | Oct 11, 2022 9:49:36 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| Asia Pacific (Singapore)  | 15.177.50.72      | Oct 11, 2022 9:49:35 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| Asia Pacific (Singapore)  | 15.177.54.70      | Oct 11, 2022 9:49:35 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| Asia Pacific (Sydney)     | 15.177.62.70      | Oct 11, 2022 9:49:41 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| Asia Pacific (Sydney)     | 15.177.58.70      | Oct 11, 2022 9:49:40 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| EU (Ireland)              | 15.177.34.70      | Oct 11, 2022 9:49:26 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| EU (Ireland)              | 15.177.38.70      | Oct 11, 2022 9:49:26 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| South America (São Paulo) | 15.177.30.70      | Oct 11, 2022 9:49:36 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| South America (São Paulo) | 15.177.26.70      | Oct 11, 2022 9:49:36 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| US East (N. Virginia)     | 15.177.6.70       | Oct 11, 2022 9:49:30 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| US East (N. Virginia)     | 15.177.2.72       | Oct 11, 2022 9:49:29 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| US West (N. California)   | 15.177.10.70      | Oct 11, 2022 9:49:35 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| US West (N. California)   | 15.177.14.70      | Oct 11, 2022 9:49:34 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |
| US West (Oregon)          | 15.177.18.70      | Oct 11, 2022 9:49:34 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.170.187.247 |
| US West (Oregon)          | 15.177.22.70      | Oct 11, 2022 9:49:34 AM UTC | Success: HTTP Status Code 200, OK. Resolved IP: 18.132.144.177 |

To view latency graphs, select a health check that has a latency graphs enabled.



Application Load Balancer



Amazon Simple Storage Service (Amazon S3)

# Routing policy

Configure two failover routing policies to record type alias and associate to health checks

Primary      Application Load Balancer (eu-west-2)  
Secondary   Amazon S3 bucket (eu-west-1)

Route 53 > Hosted zones > unicornpacket.com > Create record

**Quick create record** [Info](#) [Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#)  .unicornpacket.com Record type [Info](#)

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to [Info](#)

Alias hosted zone ID: ZHURV8PSTC4K8

Routing policy [Info](#)  Failover record type

Health check ID [Info](#)

Evaluate target health ☒ Yes

Record ID [Info](#)

[Add another record](#)

[Cancel](#) [Create records](#)



Application Load  
Balancer

# Routing policy

Configure two failover routing policies to record type alias and associate to health checks

Primary      Application Load Balancer (eu-west-2)  
Secondary   Amazon S3 bucket (eu-west-1)

Route 53 > Hosted zones > unicornpacket.com > Create record

**Quick create record** [Info](#) [Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#)  .unicornpacket.com Record type [Info](#)

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to [Info](#)

Routing policy [Info](#)  Failover record type

Health check ID - optional [Info](#)    Evaluate target health ☒ Yes

Record ID [Info](#)

[Add another record](#)

[Cancel](#) [Create records](#)



Amazon Simple Storage  
Service (Amazon S3)



# Records

Route 53 > Hosted zones > unicornpacket.com

Public unicornpacket.com Info

Delete zone Test record Configure query logging

► Hosted zone details Edit hosted zone

Records (4) DNSSEC signing Hosted zone tags (0)

Records (4) Info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

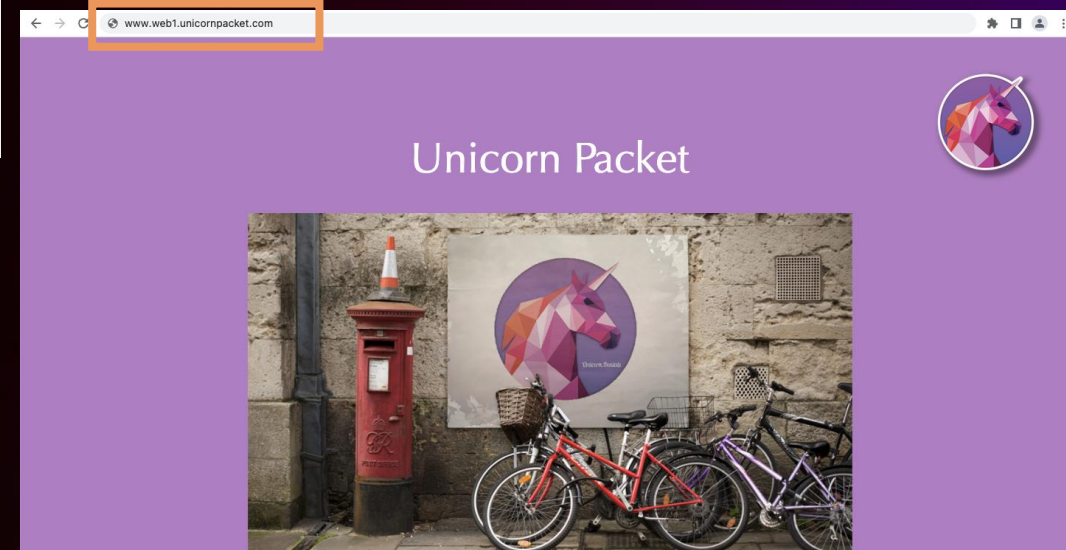
Filter records by property or value

Type Routing policy Alias

| <input type="checkbox"/> | Record name                | Type | Routin... | Differ... | Value/Route traffic to                                                                               |
|--------------------------|----------------------------|------|-----------|-----------|------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | unicornpacket.com          | NS   | Simple    | -         | ns-254.awsdns-31.com.<br>ns-1762.awsdns-28.co.uk.<br>ns-1500.awsdns-59.org.<br>ns-563.awsdns-06.net. |
| <input type="checkbox"/> | unicornpacket.com          | SOA  | Simple    | -         | ns-254.awsdns-31.com. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400                         |
| <input type="checkbox"/> | www.web1.unicornpacket.com | A    | Failover  | Primary   | dualstack.lb-unicorn-packets-52827961.eu-west-2.elb.amazonaws.com.                                   |
| <input type="checkbox"/> | www.web1.unicornpacket.com | A    | Failover  | Secondary | s3-website-eu-west-1.amazonaws.com.                                                                  |

www.web1.unicornpacket.com is now publicly accessible via Amazon Route 53

The primary record routes traffic to the Application Load Balancer eu-west-2 (London)





# Simulated failover

Simulated failure is initiated

Failover takes place and the secondary record routes traffic to the Amazon S3 bucket located in eu-west-1 (Ireland)

Health check with id 99e05ac7-075b-4106-926e-09588f84ea1c has been created successfully

Create health check Delete health check Edit health check

Filter by keyword

| Name | Status    | Description                              | Alarms                | ID                                   |
|------|-----------|------------------------------------------|-----------------------|--------------------------------------|
| web1 | Unhealthy | http://LB-Unicorn-Packets-52827961.eu... | No alarms configured. | 5091d7db-df54-4664-bf47-45948e13e3b1 |
| web2 | Healthy   | http://www.web1.unicornpacket.com.s3...  | No alarms configured. | 99e05ac7-075b-4106-926e-09588f84ea1c |

Info Monitoring Alarms Tags Health checkers Latency

View current status View last failed check Refresh

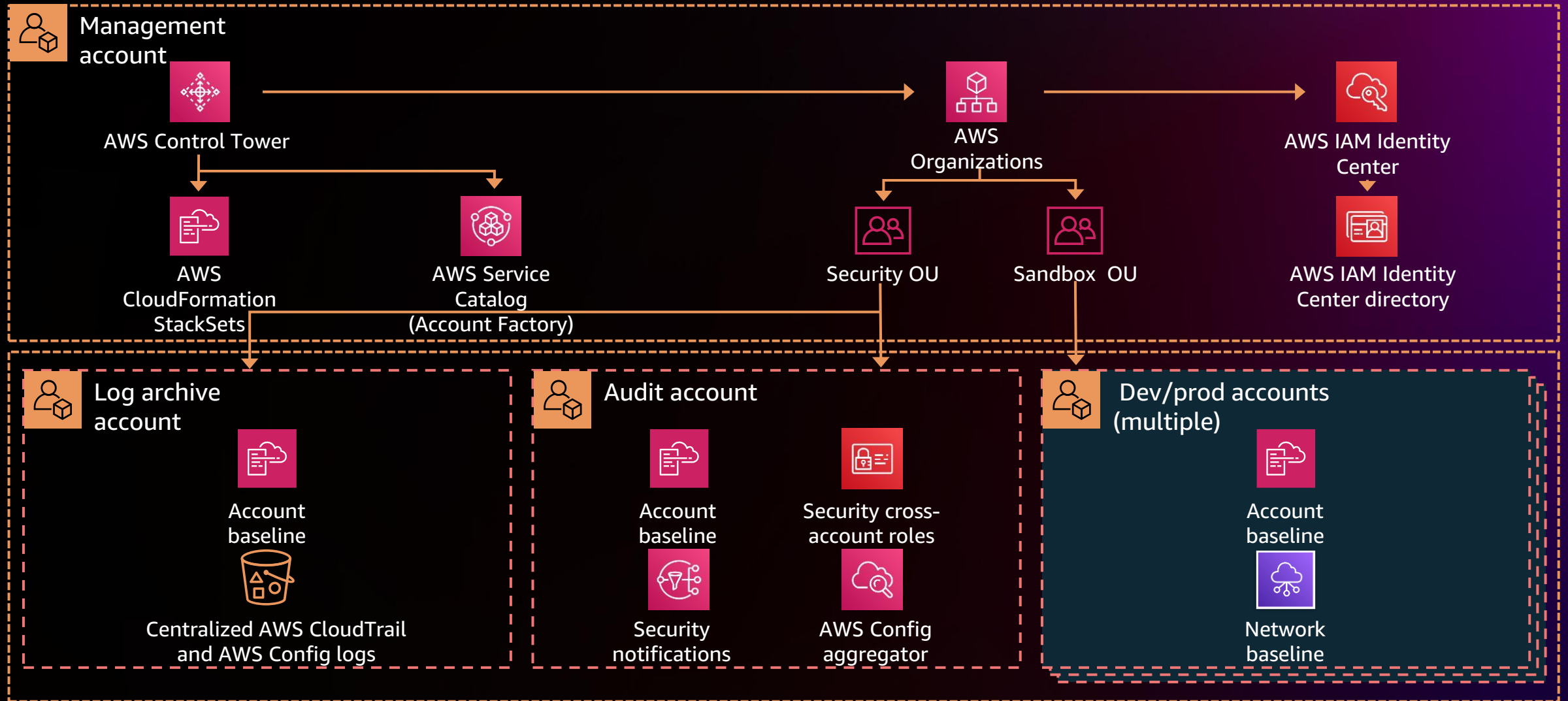
| Health checker region     | Health checker IP | Last checked                | Status                                                                |
|---------------------------|-------------------|-----------------------------|-----------------------------------------------------------------------|
| Asia Pacific (Tokyo)      | 15.177.42.70      | Oct 11, 2022 9:51:52 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| Asia Pacific (Tokyo)      | 15.177.46.70      | Oct 11, 2022 9:51:52 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| Asia Pacific (Singapore)  | 15.177.50.72      | Oct 11, 2022 9:51:51 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| Asia Pacific (Singapore)  | 15.177.54.70      | Oct 11, 2022 9:51:51 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| Asia Pacific (Sydney)     | 15.177.62.70      | Oct 11, 2022 9:51:57 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| Asia Pacific (Sydney)     | 15.177.58.70      | Oct 11, 2022 9:51:56 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| EU (Ireland)              | 15.177.34.70      | Oct 11, 2022 9:52:16 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| EU (Ireland)              | 15.177.38.70      | Oct 11, 2022 9:52:17 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| South America (São Paulo) | 15.177.30.70      | Oct 11, 2022 9:51:52 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| South America (São Paulo) | 15.177.26.70      | Oct 11, 2022 9:51:52 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| US East (N. Virginia)     | 15.177.6.70       | Oct 11, 2022 9:52:20 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| US East (N. Virginia)     | 15.177.2.72       | Oct 11, 2022 9:52:19 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| US West (N. California)   | 15.177.10.70      | Oct 11, 2022 9:51:51 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| US West (N. California)   | 15.177.14.70      | Oct 11, 2022 9:51:50 AM UTC | Failure: Resolved IP: 18.132.144.177, Connection timed out. The en... |
| US West (Oregon)          | 15.177.18.70      | Oct 11, 2022 9:51:50 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |
| US West (Oregon)          | 15.177.22.70      | Oct 11, 2022 9:51:50 AM UTC | Failure: Resolved IP: 18.170.187.247, Connection timed out. The en... |

To view latency graphs, select a health check that has latency graphs enabled.

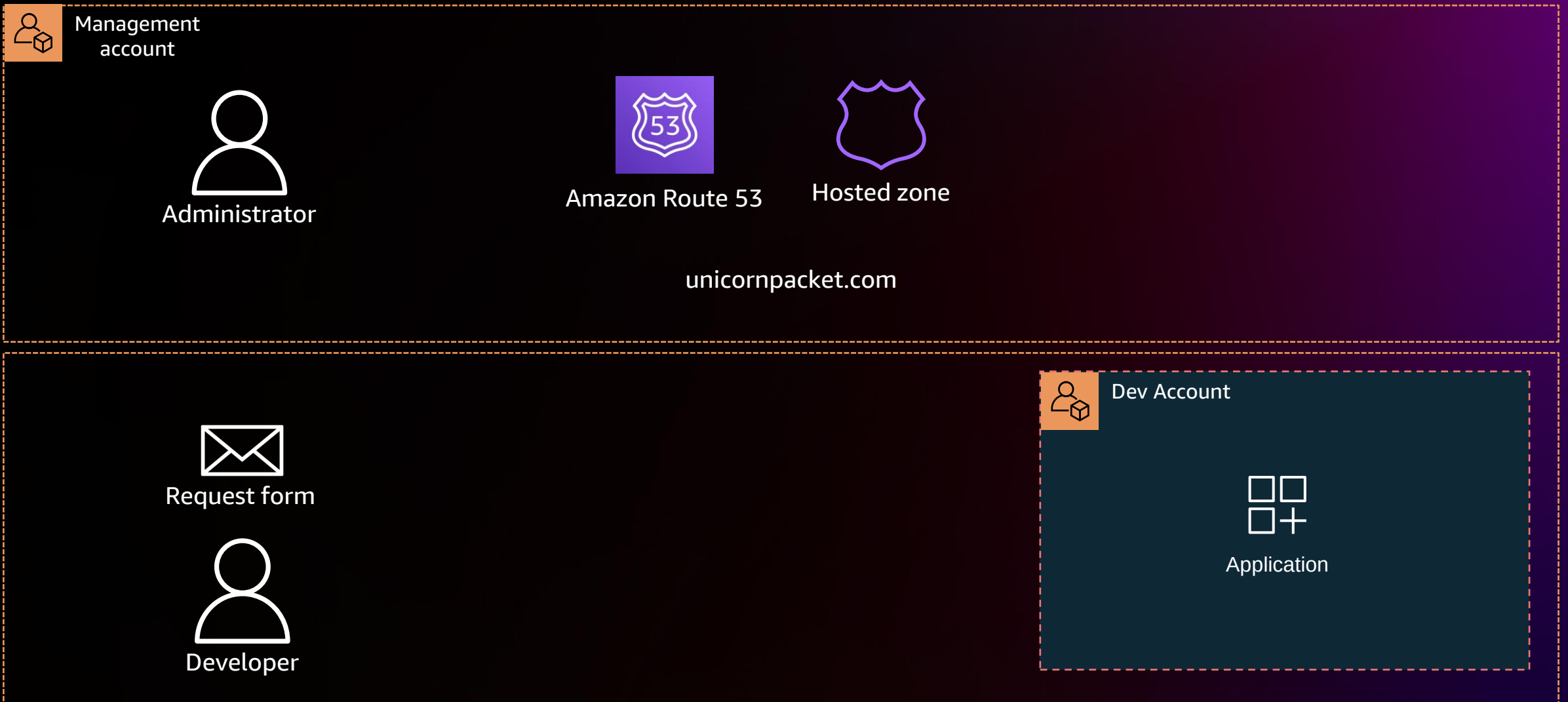


# Identity and access management

# Unicorn packet expansion



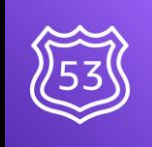
# I can't access Route 53 anymore



# I can't access Route 53 anymore!



Management  
account



Amazon Route 53



Hosted zone

unicornpacket.com

4. Create an NS Record
5. Specify name servers (step 3)

1. Create a hosted zone
2. Create record for application
3. Copy hosted zone name servers



Dev Account

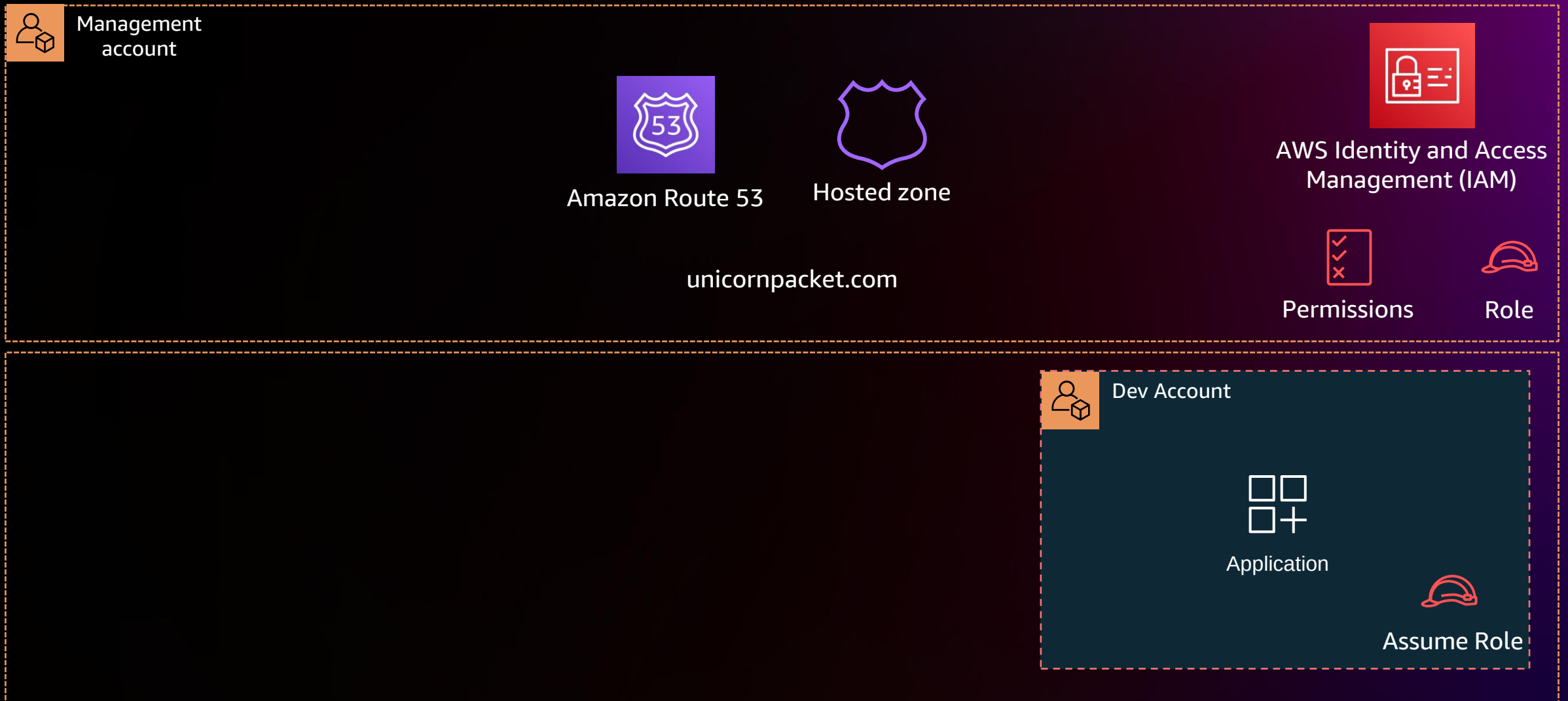


Application



Hosted zone  
dev.unicornpacket.com

# I can't access Route 53 anymore!



# AWS managed



AWS Identity and Access  
Management (IAM)



Permissions

|                                                                                                                                                                                                                                          |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|   <a href="#">AmazonRoute53DomainsReadOnlyAccess</a>               | AWS managed |
|   <a href="#">AmazonRoute53DomainsFullAccess</a>                   | AWS managed |
|   <a href="#">AmazonRoute53ReadOnlyAccess</a>                      | AWS managed |
|   <a href="#">AmazonRoute53AutoNamingReadOnlyAccess</a>            | AWS managed |
|   <a href="#">AmazonRoute53AutoNamingFullAccess</a>                | AWS managed |
|   <a href="#">AmazonRoute53AutoNamingRegistrantAccess</a>          | AWS managed |
|   <a href="#">AmazonRoute53FullAccess</a>                          | AWS managed |
|   <a href="#">Route53ResolverServiceRolePolicy</a>                 | AWS managed |
|   <a href="#">AmazonRoute53RecoveryReadinessFullAccess</a>         | AWS managed |
|   <a href="#">AmazonRoute53RecoveryControlConfigFullAccess</a>     | AWS managed |
|   <a href="#">AmazonRoute53ResolverReadOnlyAccess</a>              | AWS managed |
|   <a href="#">AmazonRoute53RecoveryControlConfigReadOnlyAccess</a> | AWS managed |
|   <a href="#">Route53RecoveryReadinessServiceRolePolicy</a>    | AWS managed |
|   <a href="#">AmazonRoute53RecoveryReadinessReadOnlyAccess</a> | AWS managed |
|   <a href="#">AmazonRoute53RecoveryClusterFullAccess</a>       | AWS managed |
|   <a href="#">AmazonRoute53ResolverFullAccess</a>              | AWS managed |
|   <a href="#">AmazonRoute53RecoveryClusterReadOnlyAccess</a>   | AWS managed |

# Create your own policy



|                                                                       |                                                                         |                                                                            |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> GetChange ?                       | <input checked="" type="checkbox"/> GetReusableDelegationSet ?          | <input checked="" type="checkbox"/> ListResourceRecordSets ?               |
| <input checked="" type="checkbox"/> GetCheckerIpRanges ?              | <input checked="" type="checkbox"/> ListCidrBlocks ?                    | <input checked="" type="checkbox"/> ListTrafficPolicies ?                  |
| <input checked="" type="checkbox"/> GetGeoLocation ?                  | <input checked="" type="checkbox"/> ListCidrCollections ?               | <input checked="" type="checkbox"/> ListTrafficPolicyInstancesByHoste... ? |
| <input checked="" type="checkbox"/> GetHealthCheckCount ?             | <input checked="" type="checkbox"/> ListCidrLocations ?                 | <input checked="" type="checkbox"/> ListTrafficPolicyInstancesByPolicy ?   |
| <input checked="" type="checkbox"/> GetHealthCheckLastFailureReason ? | <input checked="" type="checkbox"/> ListHostedZones ?                   | <input checked="" type="checkbox"/> ListTrafficPolicyVersions ?            |
| <input checked="" type="checkbox"/> GetHealthCheckStatus ?            | <input checked="" type="checkbox"/> ListHostedZonesByName ?             | <input checked="" type="checkbox"/> ListVPCAssociationAuthorizations ?     |
| <input checked="" type="checkbox"/> GetHostedZone ?                   | <input checked="" type="checkbox"/> ListHostedZonesByVPC ?              |                                                                            |
| <input checked="" type="checkbox"/> GetHostedZoneCount ?              | <input checked="" type="checkbox"/> ListQueryLoggingConfigs ?           |                                                                            |
| <input checked="" type="checkbox"/> Read (16 selected)                |                                                                         |                                                                            |
| <input checked="" type="checkbox"/> GetAccountLimit ?                 | <input checked="" type="checkbox"/> GetTrafficPolicy ?                  | <input checked="" type="checkbox"/> ListTagsForResource ?                  |
| <input checked="" type="checkbox"/> GetDNSSEC ?                       | <input checked="" type="checkbox"/> GetTrafficPolicyInstance ?          | <input checked="" type="checkbox"/> ListTagsForResources ?                 |
| <input checked="" type="checkbox"/> GetHealthCheck ?                  | <input checked="" type="checkbox"/> GetTrafficPolicyInstanceCount ?     | <input checked="" type="checkbox"/> ListTrafficPolicyInstances ?           |
| <input checked="" type="checkbox"/> GetHostedZoneLimit ?              | <input checked="" type="checkbox"/> ListGeoLocations ?                  | <input checked="" type="checkbox"/> TestDNSAnswer ?                        |
| <input checked="" type="checkbox"/> GetQueryLoggingConfig ?           | <input checked="" type="checkbox"/> ListHealthChecks ?                  |                                                                            |
| <input checked="" type="checkbox"/> GetReusableDelegationSetLimit ?   | <input checked="" type="checkbox"/> ListReusableDelegationSets ?        |                                                                            |
| <input checked="" type="checkbox"/> Tagging (1 selected)              |                                                                         |                                                                            |
| <input checked="" type="checkbox"/> ChangeTagsForResource ?           |                                                                         |                                                                            |
| <input checked="" type="checkbox"/> Write (31 selected)               |                                                                         |                                                                            |
| <input checked="" type="checkbox"/> ActivateKeySigningKey ?           | <input checked="" type="checkbox"/> CreateTrafficPolicyInstance ?       | <input checked="" type="checkbox"/> DeleteTrafficPolicyInstance ?          |
| <input checked="" type="checkbox"/> AssociateVPCWithHostedZone ?      | <input checked="" type="checkbox"/> CreateTrafficPolicyVersion ?        | <input checked="" type="checkbox"/> DeleteVPCAssociationAuthorization ?    |
| <input checked="" type="checkbox"/> ChangeCidrCollection ?            | <input checked="" type="checkbox"/> CreateVPCAssociationAuthorization ? | <input checked="" type="checkbox"/> DisableHostedZoneDNSSEC ?              |
| <input checked="" type="checkbox"/> ChangeResourceRecordSets ?        | <input checked="" type="checkbox"/> DeactivateKeySigningKey ?           | <input checked="" type="checkbox"/> DisassociateVPCFromHostedZone ?        |
| <input checked="" type="checkbox"/> CreateCidrCollection ?            | <input checked="" type="checkbox"/> DeleteCidrCollection ?              | <input checked="" type="checkbox"/> EnableHostedZoneDNSSEC ?               |
| <input checked="" type="checkbox"/> CreateHealthCheck ?               | <input checked="" type="checkbox"/> DeleteHealthCheck ?                 | <input checked="" type="checkbox"/> UpdateHealthCheck ?                    |
| <input checked="" type="checkbox"/> CreateHostedZone ?                | <input checked="" type="checkbox"/> DeleteHostedZone ?                  | <input checked="" type="checkbox"/> UpdateHostedZoneComment ?              |
| <input checked="" type="checkbox"/> CreateKeySigningKey ?             | <input checked="" type="checkbox"/> DeleteKeySigningKey ?               | <input checked="" type="checkbox"/> UpdateTrafficPolicyComment ?           |
| <input checked="" type="checkbox"/> CreateQueryLoggingConfig ?        | <input checked="" type="checkbox"/> DeleteQueryLoggingConfig ?          | <input checked="" type="checkbox"/> UpdateTrafficPolicyInstance ?          |
| <input checked="" type="checkbox"/> CreateReusableDelegationSet ?     | <input checked="" type="checkbox"/> DeleteReusableDelegationSet ?       |                                                                            |
| <input checked="" type="checkbox"/> CreateTrafficPolicy ?             | <input checked="" type="checkbox"/> DeleteTrafficPolicy ?               |                                                                            |



# New!! DNS resource record set permissions

- Specify granular IAM policies to control who can create, edit, or delete **individual resource record sets** within a hosted zone
- These granular permissions give individual DNS administrators direct ownership at the resource record set level, which may help customers avoid relying on a central team to manage changes

```
route53:ChangeResourceRecordSetsNormalizedRecordNames
route53:ChangeResourceRecordSetsRecordTypes
route53:ChangeResourceRecordSetsActions
```

# The end .... honest



# Let's summarize

- It's always DNS – in a good way
- When mistakes are made, it can be painful due to caching/TTL
- You understand how DNS works – delegation by NS record
- Domain name registration and transfer
- Hosted zones, record types, and resource record sets
- Route 53 specific capabilities – routing policies, DNSSEC, IAM

# There is more to Route 53

- Resolver endpoints
- VPC Query Logs
- DNS Firewall
- Application Recovery Controller



# Thank you!



Please complete the session survey in the **mobile app**

